

Liquid metal dispersed single-atom catalyst with high-temperature stability

Corresponding Author: Professor Lei Fu

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

The work by Ziyue Zeng's and coworkers deals with a very relevant topic in the field of catalysis. The paper is well written and contains relevant information. However, the paper is not suitable for Nature Communications and needs to be fundamentally revised in order to be suitable for a more specialised journal in the future. The reason is simple: the main claims of the article, namely that a single Pt atom can be stabilised in liquid gallium to perform high-temperature dehydrogenation catalysis in the form of supported liquid metal catalysis, are by no means new, but have already been published (with slightly different terminology) in more than 50 articles (this reviewer has verified this today using SCOPUS), including high-ranking publications in Nature Chemistry, such as the work by Rahim et al. in Nature Chemistry, Volume 14, 935-941 from 2022. (DOI: 10.1038/s41557-022-00965-6).

Even the idea to immobilize liquid Ga-Pt alloy droplets on porous supports for alkane dehydrogenation reactions is far from new. Under the name 'Supported Catalytically Active Liquid Metal Solutions (SCALMS)' (Nature Chemistry, 2017, 9(9), 862-867, DOI: 10.1038/nchem.2822, and paper citing this work) more than 30 papers have been published describing exactly this concept, 11 of these deal with Ga-Pt systems including preparation by ultrasonication and theoretical calculations to prove the isolated nature of Pt in Ga. Seven of these papers deal with propane dehydrogenation which is a very similar reaction to ethane hydrogenation.

In this light, this reviewer finds it striking that none of these earlier works were cited by Ziyue Zeng and coworkers. This is strange – and even if one takes into account the slightly different terminology used in the various articles – a proper and rigorous approach could have led to the conclusion that the work described in the current submission lacks the required novelty for a publication in Nature Communications.

In conclusion, this reviewer recommends studying the existing literature much more carefully and then publishing some of the interesting detailed information in the submitted manuscript in a much more specialised journal.

Reviewer #2

(Remarks to the Author)

In the manuscript, the authors described a strategy to realize the highly thermal-stable single-atom catalysts (SACs) resistance to sintering through harnessing the inherent metal affinity and fluidity of liquid metal. This work provides a novel approach for addressing the thermal instability of SACs under high-temperature conditions, which is regarded as the most critical and persistent challenge in single-atom catalysis. The proposed liquid metal-dispersed SACs exhibited outstanding thermal durability for ethane dehydrogenation, sustaining stable operation for over 100 h at 650 °C with an impressive ethylene selectivity of 98%. This work not only resolves a key limitation in SACs but also paves the avenue to practical deployment of SACs for practical industrial thermocatalysis.

Hence, I would like to recommend the publication in Nature Communications. The following are several minor details for the authors to consider:

1. It is very interesting to use liquid metal with dynamic fluidity and metallicity to stabilize the single atoms. How can a single atom be "anchored" on this dynamically renewed surface? Is this anchoring a static "locking" or a dynamic "stabilization" that relies on the rapid dynamic reconfiguration of the liquid metal surface? I'd be very glad to hear some explanations or assumptions.

2. As demonstrated in page 8 line 6, the Pt–Pt bond length is 2.65~2.79 Å. However, in Figure 2d, the authors drew a gray area between 2.65~3.35 Å. Does the grey area represent the inherent bond length of Pt–Pt, or is there any specific basis or standard for this range?

3. In Figure 2f, the potential energy of Pt–Pt (107.9 kcal/mol) is lower than that of Ga–Pt and Ga–Ga. I wonder whether this is in contradiction to the opinion that Pt tends to form Ga–Pt bonds within liquid metal.

4. Pt is regarded as a good catalyst for ethane dehydrogenation. I am very curious whether Ga and porous MFI zeolite (ZSM-5) will have any impact on the conversion and selectivity of the ethane dehydrogenation reaction. Ga/ZSM-5 was also proposed effective for dehydrogenation of alkanes (eg. DOI:10.1016/j.jcat.2024.115834; 10.1021/acscatal.7b04295, etc.). It would be better for the authors to elaborate on the catalytic performances of Pt/ZSM-5 and Pt@Ga/ZSM-5.

5. The authors have demonstrated the unique advantages of liquid metal in dispersing single atoms and maintaining their high-temperature stability. I am curious whether liquid metal has any advantages over the solid carriers for synthesizing SACs. Please give corresponding explanations.

6. A few specific minor errors:

1) Redefining abbreviations: The authors have already defined the single-atom catalysts and given the abbreviation in line 2 of the abstract, but it is abbreviated again in line 5.

2) Spelling mistake: in Supplementary Fig. 1, the word “luctuates” may be misspelled.

3) Other error: in Supplementary Fig. 11, the x-coordinate is incorrect.

Version 1:

Reviewer comments:

Reviewer #2

(Remarks to the Author)

I read the manuscript carefully and found that the authors did a good job of revising it. The new version looks much better compared to the original one. As my concerns were fully addressed in this revision, I believe the manuscript is now publishable.

Version 2:

Reviewer comments:

Reviewer #4

(Remarks to the Author)

In the manuscript entitled “Liquid metal dispersed single-atom catalyst with high-temperature stability” submitted to Nat. Commun., Zeng et al. report a Pt-based single-atom catalyst dispersed in a Ga-based molten phase and supported on ZSM-5 for ethane dehydrogenation to ethylene. The authors imply that this constitutes a new conceptual way of stabilizing SACs

with high activity and thermal stability, and they support their arguments by a combination of experimental data and AIMD calculations.

My main concern is that the central experimental claim about the unique role of Pt is not convincingly supported by the data as described. The authors state in their rebuttal letter that, upon “comparing the conversion and selectivity of pure ZSM-5, pure Ga, and Pt@Ga/ZSM-5, it is obvious that both ZSM-5 and Ga have no impact on the activity.” This is difficult to reconcile with the reported performance: Ga metal alone reaches approximately half the conversion of the Pt–Ga single-atom system and even exhibits higher selectivity to ethylene. These numbers strongly suggest that Ga is an active and selective catalyst on its own and that the incremental contribution of Pt to the overall performance is limited. Rather than demonstrating a clear, dominant role of Pt single atoms in a molten Ga host, the data point to a system in which Ga already performs well and Pt provides a modest, enhancement in conversion at (perhaps) the cost of selectivity. The authors need to acknowledge and analyze this more carefully; in its current form, the narrative that Ga and ZSM-5 “have no impact on the activity” is not credible.

A second major issue is the way the work is positioned with respect to existing literature on SCALMS (Supported Catalytically Active Liquid Metal Solutions) and single-atom/alloy catalysts. Ga–Pt and related molten metal/solid alloy catalysts have been studied for some time, in particular by Peter Wasserscheid and co-workers and others, who have explicitly discussed high-temperature stability and among others Ga-based systems. There are already reports that can reasonably be described as “single atom” or “single site” catalysts embedded in a liquid Ga host on a support, including recent work (e.g. doi:10.1002/cphc.202400651) that addresses exactly the stability and site isolation arguments the authors here put forward. Against this background, the assertion that this manuscript is the first to propose the concept of a supported single-atom catalyst in a molten metal host appears overstated. The results read as an incremental variation within the established SCALMS/SAA/SAC framework rather than as a fundamentally new conceptual advance.

In addition, the experimental scope is rather narrow for a paper targeting Nature Communications. The study focuses on a single reaction (ethane to ethylene) under relatively limited conditions, with a small set of catalyst formulations and without an in-depth kinetic analysis (e.g. an estimate of turnover frequencies per Pt vs. Ga, reaction orders) or operando characterization that would firmly establish the structure and environment of the proposed Pt single atoms under working conditions. Long-term stability, deactivation pathways, and structure–performance relationships are not explored to the depth one would expect for the broad readership of such a journal. The resulting dataset is essentially a proof of concept confined to one reaction and one composition space, which may be more appropriate for a specialized catalysis journal.

In summary, the manuscript is methodologically solid and the topic is of interest to the catalysis community, but the incremental nature of the contribution, the limited experimental scope, the overstatement of conceptual novelty compared to the established SCALMS and SAC/SAA literature, and the unconvincing treatment of the role of Pt versus Ga prevent me from viewing this as a strong candidate for publication in Nat. Commun. I therefore do not recommend publication of this manuscript in its current form.

Reviewer #5

(Remarks to the Author)

This manuscript reports a liquid metal catalytic system in which Pt dissolved in liquid Ga exhibits exceptionally stable performance in ethane dehydrogenation. Although the concept of using liquid metals in catalysis is not entirely new, the remarkable long-term stability demonstrated here is still appealing. The theoretical analysis, especially the MD simulations, on the thermodynamic stability of the single-atom Pt species is also insightful. In my view, this work is suitable for publication in Nature Communications.

However, a few points require clarification.

1. In Fig. 2e, the Pt–Pt interaction appears stronger than that of Pt–Ga. Why, then, does platinum remain atomically dispersed in liquid gallium? The authors argue that “The calculated bond dissociation energies show a clear ranking: $E(\text{Pt–Pt})$ (107.9 kcal/mol) > $E(\text{Ga–Pt})$ (74.9 kcal/mol) > $E(\text{Ga–Ga})$ (32.5 kcal/mol). Crucially, the formation of two Ga–Pt bonds (149.8 kcal/mol) is energetically favored over either maintaining a Pt–Pt bond (107.9 kcal/mol) or the combined Pt–Pt and Ga–Ga bonding (140.4 kcal/mol).” Yet, this explanation is not entirely convincing, because the numbers of the formed and dissociated bonds seem over simplified for the liquid phase.

2. On the other hand, the B3LYP functional is known to be less accurate for transition metal dimers due to multi-configurational effects (see relevant references: Alvaro Posada-Borbón, Alvaro Posada-Amarillas, Theoretical DFT study of homonuclear and binary transition-metal dimers, Chemical Physics Letters, 2015; Haoyu S. Yu, Xiao He, Donald G. Truhlar, MN15-L: A New Local Exchange–Correlation Functional for Kohn–Sham Density Functional Theory with Broad Accuracy for Atoms, Molecules, and Solids, J. Chem. Theory Comput. 2016.) Given this, are the results in Fig. 2(e) sufficiently robust?

3. Additionally, what is the turnover frequency (TOF) normalized by the mass of Pt? This information is crucial for evaluating the atomic efficiency and cost-effectiveness of the catalyst, given the high price of platinum?

Version 3:

Reviewer comments:

Reviewer #4

(Remarks to the Author)

The authors have responded to all the reviewers' comments appropriately.

Reviewer #5

(Remarks to the Author)

The authors have satisfactorily addressed all of my previous concerns. I recommend accepting the manuscript in its present form.

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Reply to the referees

To Referee #1

First of all, we really appreciate your constructive comments, which significantly promote the improvement of our work. According to your suggestions, we have made corresponding revisions to our manuscript as listed below.

Overall remark: *The work by Ziyue Zeng's and coworkers deals with a very relevant topic in the field of catalysis. The paper is well written and contains relevant information. However, the paper is not suitable for Nature Communications and needs to be fundamentally revised in order to be suitable for a more specialised journal in the future. The reason is simple: the main claims of the article, namely that a single Pt atom can be stabilised in liquid gallium to perform high-temperature dehydrogenation catalysis in the form of supported liquid metal catalysis, are by no means new, but have already been published (with slightly different terminology) in more than 50 articles (this reviewer has verified this today using SCOPUS), including high-ranking publications in Nature Chemistry, such as the work by Rahim et al. in Nature Chemistry, Volume 14, 935-941 from 2022. (DOI: 10.1038/s41557-022-00965-6).*

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In this light, this reviewer finds it striking that none of these earlier works were cited by Ziyue Zeng and coworkers. This is strange – and even if one takes into account the slightly different terminology used in the various articles – a proper and rigorous approach could have led to the conclusion that the work described in the current submission lacks the required novelty for a publication in Nature Communications.

In conclusion, this reviewer recommends studying the existing literature much more carefully and then publishing some of the interesting detailed information in the submitted manuscript in a much more specialised journal.

Author reply: Thanks for your comment. We have made a point-by-point response to your comments and carefully revised the manuscript. For your reference, please find our revisions marked in the red copy. We hope the newly revised manuscript can be more understandable and informative to meet the demand for publication in *Nature Communications*.

1) In terms of the query that a single Pt atom can be stabilized in liquid gallium, the reviewer considered that it has been reported. We would like to clarify.

Our work focuses on the severe sintering limitation of SACs at high temperatures in industrial catalysis. By leveraging the fluidity and metal affinity of liquid metal, we realized the ultra-stable SACs at 650 °C. It is worth noting that the high temperature contributes to the dispersion of single atoms in our work rather than the agglomeration sintering phenomenon in reported works, which is of great significance for promoting the industrial application of SACs. The significance of our work lays as follows:

i) **The realization of highly thermal-stable SACs for industrial catalysis.** SACs have emerged as promising candidates to replace conventional industrial catalysts owing to their exceptional catalytic activity and atomic efficiency. The high surface energy of single atoms renders them prone to migration and subsequent sintering during catalytic processes, especially at high temperatures. Thus, it remains **a formidable obstacle to developing novel catalysts with high-temperature thermal stability**. We realized the highly thermal-stable SACs resistance to sintering at harsh conditions through harnessing the inherent metal affinity and fluidity of liquid metal. We believe this is a crucial step forward in promoting the application of SACs in high-temperature industrial catalysis.

ii) **A brand-new strategy for achieving highly thermal-stable SACs resistance to sintering by liquid metals.** The monodispersion of Pt in liquid metals rather than agglomeration at high temperatures is a thermodynamically favorable process, which is owing to the selective breaking and formation of chemical bonds profiting from the inherent metal affinity of liquid metal. Combined with the fluidity of liquid metal, active metal atoms can move but remain confined to the liquid metal as the metallic single-atom state at high temperatures. We have provided robust evidence for clarifying the mechanism.

Regarding the work by Rahim et al. in *Nature Chemistry*, Volume 14, 935–941 from 2022 (DOI: 10.1038/s41557-022-00965-6), we acknowledge that this work did indeed disperse ~0.0001 at% Pt in liquid metal and applied it to the reactions that are limited at room temperature, including the oxidation of pyrogallol, the oxidation of 3,3',5,5'-tetramethylbenzidine, the reduction of methylene blue, and the electrochemical oxidation

of CH₃OH. However, this article (*Nat. Chem.* **2022**, *14*, 935) does not suggest that liquid metal carriers can solve the problem of agglomeration of single atoms under high-temperature conditions.

Therefore, the **core innovation of work and the previously reported works are essentially different**. We don't think these two articles conflict.

2) As for immobilizing liquid Ga–Pt alloy droplets on porous supports, it has **no relationship with the core innovation** of our work. The focus in this work is to utilize liquid metal to address the bottleneck of agglomeration of SACs.

Firstly, the adoption of porous support is to **reveal the thermal-stability essence for the SACs dispersed by liquid metals while eliminating the influence of mass transfer**. Porous support is a commonly used strategy to enhance the mass transfer, as it guarantees a relatively chemical inertness and the absence of mass-transfer restrictions because of the large pore size and rich pore structure (*Nat. Chem.* **2017**, *9*, 862). We have already clarified this in our manuscript (Fig. R1).

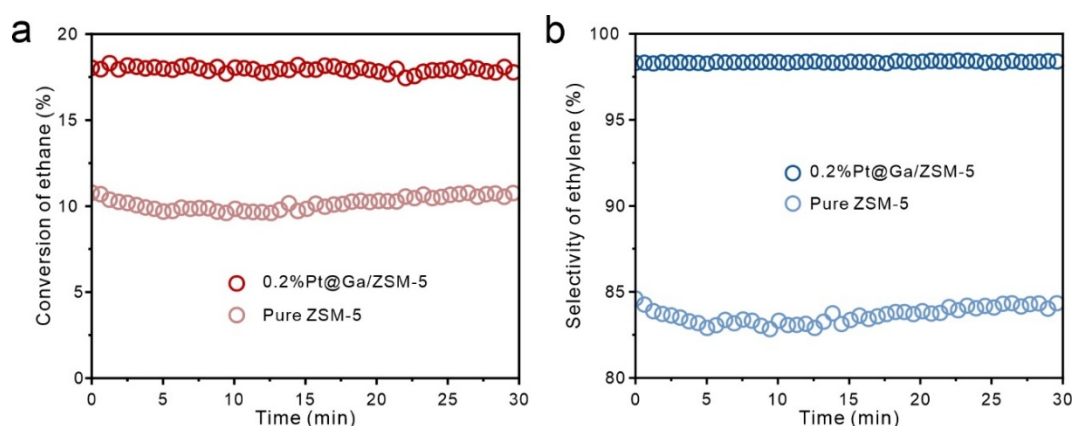


Fig. R1 | Ethane dehydrogenation performance of pure ZSM-5 and Pt@Ga/ZSM-5.

Secondly, in response to your doubts, we have conducted a comprehensive literature search via SCOPUS, retrieving studies on Ga–Pt alloys for alkane dehydrogenation and identifying five studies that dispersed the alloy on porous supports. In all these works, the porous carrier served merely to enhance mass transfer, and the main focus of the research lies in the catalytic performance of the Ga–Pt alloy itself. The work (*Nat. Chem.* **2017**, *9*, 862) highlighted by the reviewer is the first to introduce liquid metal into traditional supported catalysts. It involves the synthesis of a Ga–Pd bimetallic alloy by dissolving Pd within Ga and loading the alloy on the porous glass. And the authors employed Ga–Pd alloy for propane dehydrogenation, which manifested ~50% selectivity to butenes at 445 °C for

25 h. Compared to this work, our work demonstrates superiority from both material design and catalytic performance perspectives. **We realized highly stable SACs with uniformly coordinated active centers that significantly enhance selectivity. When applied to high-temperature ethane dehydrogenation, the catalyst maintains 98% selectivity at 650 °C for 100 h, substantially outperforming the referenced work.** Furthermore, we have provided theoretical evidence and detailed structural characterization to explore the mechanism of the excellent thermal stability of the catalyst. And for large-scale industrial catalysis, the high cost of Pt significantly constrains large-scale applications. While existing research attempts to reduce Pt usage through alloying with Ga, achieving a balance between platinum dispersion and activity in ethane dehydrogenation remains challenging. Furthermore, alkane dehydrogenation involves multi-step reaction pathways (such as dehydrogenation, desorption, and re-adsorption), and competitive reactions such as dehydrogenation, hydrogenation, and cracking may simultaneously occur on the Pt catalyst surface. Consequently, the synthesis of Pt-based catalysts with highly uniform coordination environments still presents significant challenges. In this regard, our work provides an atom-economical solution for practical industrial catalysis.

3) Regarding the reviewer's comment about the omission of relevant references, we have conducted a comprehensive literature search via SCOPUS as suggested. Our work focuses on addressing the critical sintering challenges in SACs toward high-temperature industrial catalysis. Consequently, the cited references predominantly described the current state and development of the SACs in our previous manuscript. Following your valuable suggestion, we have now incorporated additional references on liquid metal-based catalysis to make the background introduction more comprehensive. We remain confident that our work demonstrates significant innovation compared to previous studies in the field.

Changes: Thanks for your comment. We acknowledged the previous research on liquid metals conducted by others and have included references to it in the text. For your reference, we have added the corresponding references in our revised manuscript. Please find our revisions marked in the red copy. Finally, we thank the reviewer's suggestion to make our manuscript more rigorous, and we hope these revisions make our manuscript more understandable and informative to readers. Please find our revisions marked in the red copy of the revised manuscript.

21 Taccardi, N. *et al.* Gallium-rich Pd–Ga phases as supported liquid metal catalysts. *Nat. Chem.* **9**, 862–867 (2017).

22 Rahim, M. A. *et al.* Low-temperature liquid platinum catalyst. *Nat. Chem.* **14**, 935–941 (2022).

- 23 Tang, J. *et al.* Dynamic configurations of metallic atoms in the liquid state for selective propylene synthesis. *Nat. Nanotech.* **19**, 306–310, (2023).
- 35 Sattler, J. J. H. B., Ruiz-Martinez, J., Santillan-Jimenez, E. & Weckhuysen, B. M. Catalytic dehydrogenation of light alkanes on metals and metal oxides. *Chem. Rev.* **114**, 10613–10653 (2014).
- 36 Zhang, T. *et al.* Synergistic mechanism of platinum-GaO_x catalysts for propane dehydrogenation. *Angew. Chem. Int. Ed.* **61**, e202201453 (2022).
- 37 Newmeyer, E.-R. *et al.* Plasmonic charge localization and C–H activation at single-atom sites in dilute copper platinum alloys. *J. Am. Chem. Soc.* **147**, 11789–11799 (2025).
- 38 Searles, K. *et al.* Highly productive propane dehydrogenation catalyst using silica-supported Ga–Pt nanoparticles generated from single-sites. *J. Am. Chem. Soc.* **140**, 11674–11679 (2018).
- 39 Cheng, Q. *et al.* Highly efficient catalytic propane dehydrogenation driven by MFI zeolite defect sites. *Nat. Commun.* **16**, 5696 (2025).

To Referee #2

We appreciate the referee's positive evaluation of our manuscript. We have made a point-by-point response to your comments and carefully revised the manuscript as you suggested. For your reference, please find our revisions marked in red.

Overall remark: *In the manuscript, the authors described a strategy to realize the highly thermal-stable single-atom catalysts (SACs) resistance to sintering through harnessing the inherent metal affinity and fluidity of liquid metal. This work provides a novel approach for addressing the thermal instability of SACs under high-temperature conditions, which is regarded as the most critical and persistent challenge in single-atom catalysis. The proposed liquid metal-dispersed SACs exhibited outstanding thermal durability for ethane dehydrogenation, sustaining stable operation for over 100 h at 650 °C with an impressive ethylene selectivity of 98%. This work not only resolves a key limitation in SACs but also paves the avenue to practical deployment of SACs for practical industrial thermocatalysis. Hence, I would like to recommend the publication in Nature Communications.*

Author reply: We appreciate your positive evaluation of our manuscript to be published in *Nature Communications*. We have made a point-by-point response to your comments and carefully revised the manuscript as you suggested. For your reference, please find our revisions marked in the red copy of our revised manuscript.

Comment 1: *It is very interesting to use liquid metal with dynamic fluidity and metallicity to stabilize the single atoms. How can a single atom be "anchored" on this dynamically renewed surface? Is this anchoring a static "locking" or a dynamic "stabilization" that relies on the rapid dynamic reconfiguration of the liquid metal surface? I'd be very glad to hear some explanations or assumptions.*

Author reply: Thank you for your valuable comment. The anchoring of single atoms (Pt) in liquid metals fundamentally relies on a dynamic equilibrium between fluidity and stability, achieved through the synergistic interplay of stable metal bonds and surface reconfiguration. Profiting from the superior metal affinity of Ga with Pt, Pt–Pt bonds are broken at elevated temperatures, and subsequently the mobile Pt atoms are captured by Ga to form a stable Pt–Ga interaction, accomplishing the highly efficient dispersion of Pt single atoms. The fluidity of liquid metal and the stable Pt–Ga interaction ensure that the Pt atoms can remain dynamically scattered throughout the whole liquid metal dispersion system.

To further validate the stability mechanism of Pt single atoms dispersed in Ga, we constructed a $\text{Ga}_{79}\text{Pt}_1$ slab model. AIMD simulations were conducted to investigate the motion trajectory of a single Pt atom in molten Ga (analyzing its vertical distance to the surface). Throughout the 10 ps simulation period, a single Pt atom migrates relatively freely within molten Ga, with a depth from the Ga surface ranging from 0 to 4.84 Å (Fig. R2). Statistical analysis indicates a surface exposure probability of 56.3% for the isolated Pt atom. Furthermore, considering the susceptibility of metallic Ga to oxidation state, even under reducing atmospheres (e.g., H_2) at 923 K, a thin Ga_2O_3 layer may persist. To eliminate the influence of Ga oxidation on Pt exposure, we also constructed a $\text{Ga}_{79}\text{Pt}_1\text{O}_{10}$ slab model incorporating a small number of oxygen atoms and performed a 10 ps AIMD simulation. The simulation results demonstrate that limited oxygen atoms form Ga_2O_3 clusters with surface Ga atoms. Under the influence of these Ga_2O_3 clusters at the molten Ga surface, the Pt atom primarily remains within the surface and subsurface regions, exhibiting a depth range of 0 to 2.97 Å (Fig. R2). Notably, the surface exposure probability of the single Pt atom increases significantly to 81.3%. The single Pt atom in liquid metal migrates freely as Ga_8Pt or Ga_9Pt clusters, retaining a finite probability of surface exposure. Crucially, the presence of Ga_2O_3 clusters at the molten Ga surface dramatically enhances the probability of single Pt atom surface exposure. This indicates that the **Pt single atom can be stabilized to the molten Ga surface at 923 K over extended periods, thereby serving as active sites for the catalytic activation of C_2H_6 molecules.**

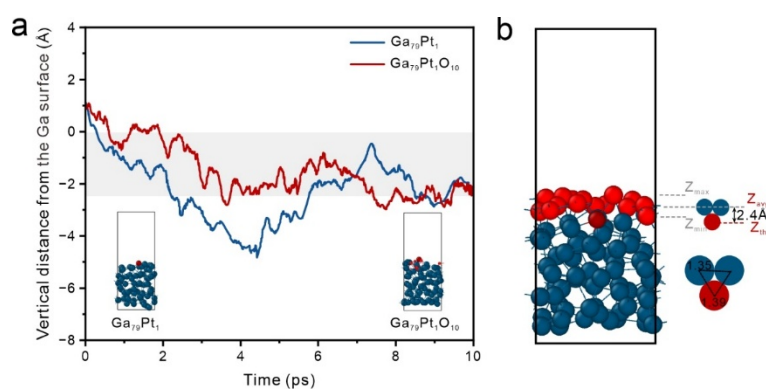


Fig. R2 | The calculated curve of the distance between a single Pt atom and the surface Ga layer as a function of time. Here, surface Ga atoms are characterized as top-layer Ga atoms in each frame structure whose coordination number is two less than the maximum coordination number, with the zero point in the z-direction defined as the average of the maximum and minimum z-coordinates of these surface Ga atoms. Furthermore, the z-coordinate threshold for identifying exposed Pt atoms is defined as the vertical distance corresponding to the closest packing radius of Re atoms relative to this zero reference point.

Therefore, the anchoring of a single atom does not rely on static “rigid locking” but is a dynamic process.

Comment 2: *As demonstrated in page 8 line 6, the Pt–Pt bond length is 2.65~2.79 Å. However, in Figure 2d, the authors drew a gray area between 2.65~3.35 Å. Does the grey area represent the inherent bond length of Pt–Pt, or is there any specific basis or standard for this range?*

Author reply: Thank you for your kind reminding. The gray area is marked to display the range of bond length fluctuation during bonding between two Pt atoms. Beyond this threshold, the Pt–Pt bond is considered dissociated, with the atoms existing in a single-atom state. The intrinsic bond length of Pt–Pt fluctuates between 2.65 and 2.79 Å (*J. Am. Chem. Soc.*, 1978, 100, 2074–2079). And generally, when the distance between the two atoms is less than 1.2 times ($2.79 \text{ Å} \times 1.2 = 3.35 \text{ Å}$) of the sum of their Cambridge Structural Database (CSD) covalent radii, it can be considered that no chemical bond has been formed between these two atoms and they can be regarded as independent of each other (*J. Comput. Chem.*, **2011**, 33, 580; *J. Chem. Phys.*, **2024**, 161, 082503). Thus, the gray area spanning from 2.65 to 3.35 Å was considered to be the distance of the Pt–Pt bond. And we have clarified this point in our revised manuscript in Figure 2 (Supplementary Information).

Comment 3: *In Figure 2f, the potential energy of Pt–Pt (107.9 kcal/mol) is lower than that of Ga–Pt and Ga–Ga. I wonder whether this is in contradiction to the opinion that Pt tends to form Ga–Pt bonds within liquid metal.*

Author reply: Thank you for your kind reminding. As you put forward, the potential energy of Pt–Pt (107.9 kcal/mol) is lower than that of Ga–Pt (74.9 kcal/mol) and Ga–Ga (32.5 kcal/mol) (Fig. R3). However, the dispersion of Pt in liquid metal Ga involves two processes, including the breaking of the Pt–Pt bond and the Ga–Ga bond, and the formation of two Ga–Pt bonds (149.8 kcal/mol). This is an exothermic process (9.4 kcal/mol). Moreover, the Ga–Ga bond in liquid Ga is weaker than that in Ga₂ molecules. Therefore, the breaking of Pt–Pt bonds to form Ga–Pt bonds is more favorable thermodynamically, which is consistent with the perspective that Pt readily forms Ga–Pt bonds in liquid metal Ga. The energetic advantage accompanied by the good affinity between Ga and Pt enables Ga atoms to disrupt Pt–Pt bonds to form stable Pt–Ga–Pt bridging units. We also provided a detailed elaboration on this matter from the perspectives of thermodynamics and kinetics in the manuscript.

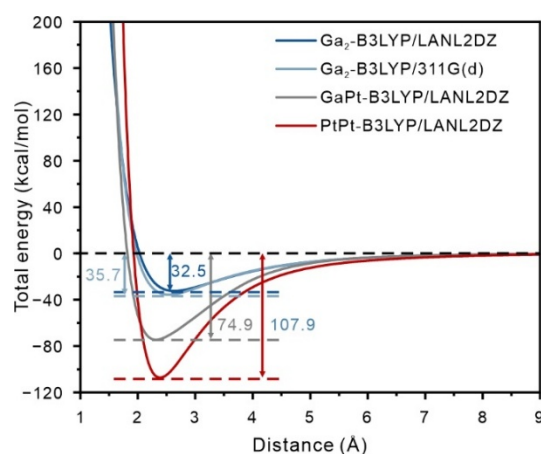


Fig. R3 | The calculated potential energy curves of three diatomic systems: Ga₂, GaPt, and Pt₂.

Comment 4: *Pt is regarded as a good catalyst for ethane dehydrogenation. I am very curious whether Ga and porous MFI zeolite (ZSM-5) will have any impact on the conversion and selectivity of the ethane dehydrogenation reaction. Ga/ZSM-5 was also proposed effective for dehydrogenation of alkanes (eg. DOI:10.1016/j.jcat.2024.115834; 10.1021/acscatal.7b04295, etc.). It would be better for the authors to elaborate on the catalytic performances of Pt/ZSM-5 and Pt@Ga/ZSM-5.*

Author reply: Thank you for your kind reminding. In our work, Ga was adopted as the dispersion to achieve atomic dispersion of the active metal. This ensures a highly uniform coordination environment of Pt atoms, which contributes to the **selectivity and durability of the SACs**. The introduction of zeolite is a commonly used strategy to **enhance mass transfer**. The adoption of Porous MFI zeolite (ZSM-5) is to reveal the thermal-stability essence for the SACs dispersed by liquid metals while eliminating the influence of mass transfer. We have supplemented the catalytic performance data of Pt/ZSM-5 and Pt@Ga/ZSM-5 to clarify the roles of the liquid metal and ZSM-5, hoping to address your questions.

1) Ga facilitates the atomic dispersion of active metals, and the highly uniform and stable coordination environment of active centers **underpins the exceptional selectivity and durability of the SACs**. Pt is recognized as an effective catalyst for alkane dehydrogenation (*Angew. Chem. Int. Ed.* **2024**, 63, e202409556). However, its catalytic activity varies significantly across different particle sizes with a varied coordination environment of each Pt atom (*ACS Catal.* **2025**, 15, 12420; *ACS Catal.* **2021**, 11, 9279). When used directly in alkane dehydrogenation, this anisotropy promotes side reactions and reduces overall selectivity. By dispersing Pt as single atoms onto Ga, a highly uniform coordination

environment is established, which ensures superior catalytic selectivity. Besides, the catalytic performances of Pt/ZSM-5 and Pt@Ga/ZSM-5 were investigated to further demonstrate the role of liquid metal in enhancing catalytic performance. As shown in Fig. R4, benefiting from the introduction of Ga, the reaction conversion of ethane is increased from 10% to 18%, while the selectivity of ethylene is improved from 82% to 98%.

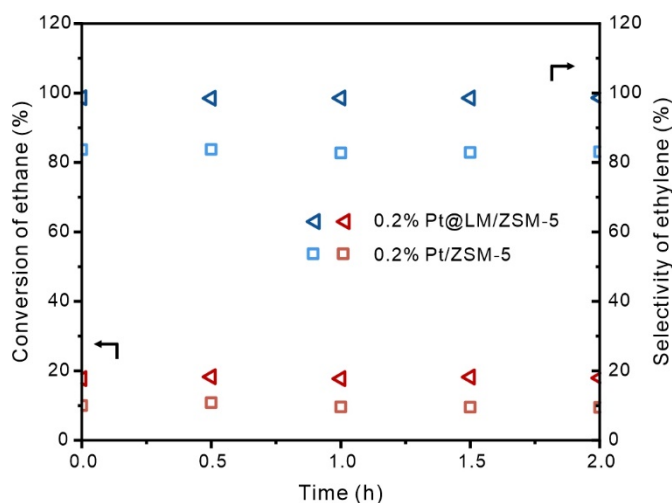


Fig. R4 | Ethane dehydrogenation performance of Pt/ZSM-5 and Pt@Ga/ZSM-5.

2) The molecular sieve itself exhibits negligible catalytic activity, with a selectivity of only 85% (Fig. R5). Its incorporation is intended to enhance gas-catalyst contact and improve mass transfer efficiency.

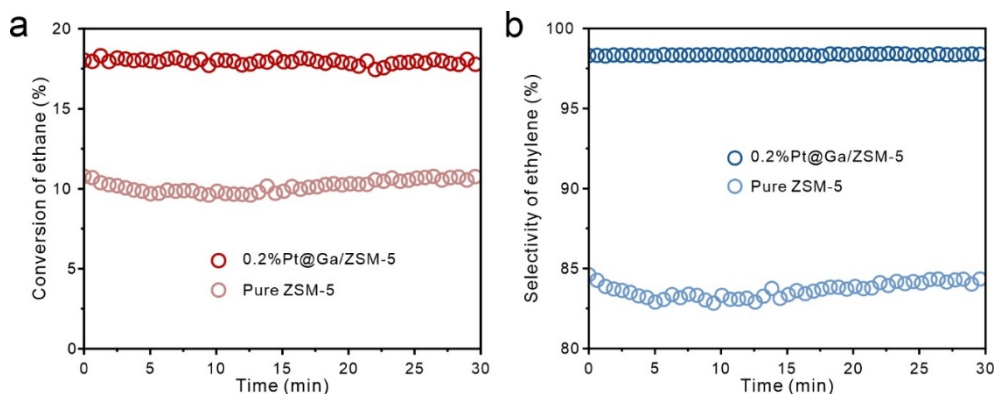


Fig. R5 | Ethane dehydrogenation performance of pure ZSM-5 and Pt@Ga/ZSM-5.

Comment 5: The authors have demonstrated the unique advantages of liquid metal in dispersing single atoms and maintaining their high-temperature stability. I am curious whether liquid metal has any advantages over the solid carriers for synthesizing SACs. Please give corresponding explanations.

Author reply: Thank you for your constructive comment. The **traditional synthesis strategies** of SACs mainly contain impregnation method (*J. Am. Chem. Soc.* **2025**, *147*, 32729), mass-selected soft-landing method (*ACS Nano* **2020**, *14*, 13279), coprecipitation method (*J. Am. Chem. Soc.* **2025**, *147*, 18737), atomic layer deposition strategy (*Nature* **2025**, *640*, 668), the host-guest strategy (*Nat. Chem.* **2020**, *12*, 764), photochemical synthesis (*Science* **2016**, *352*, 797), etc. In most of these methods, metal precursors (such as metal nitrates, metal chlorides, and metal acetylacetonates) are selected as the starting materials and subjected to heat treatment or light irradiation to deconstruct the original structure (*J. Am. Chem. Soc.* **2022**, *144*, 18155). Traditional synthesis methods of SACs require intricate procedures, including impregnation, aging, multiple drying cycles, temperature-controlled thermal treatment, and acid washing purification. These processes are not only **lengthy** but also **involve multiple sensitive variables** (e.g., pH, temperature, heating rate), posing significant challenges to reproducibility. Furthermore, these methods **usually cannot be simply extrapolated to other systems** due to their high dependence on metal-support interactions.

The advantages of liquid metal for synthesizing SACs are as follows:

1) The convenience and highly reproducible ability. The conventional synthesis strategies necessitate precise sequential control over impregnation, drying, pyrolysis, and acid washing. In contrast, the liquid metal strategy requires only physical mixing of active metal precursors (e.g., nanoparticles or even foils) with low-melting-point liquid metals (e.g., gallium, indium), followed by heating above the melting point under an inert atmosphere. Active metals spontaneously dissolve and disperse into single atoms within the liquid metal dispersion, eliminating all intermediate processing steps.

2) Excellent universality. Traditional synthesis approaches suffer inherent limitations due to their reliance on metal-support interactions, which cannot be simply extrapolated to other systems. Typically, one approach is only applicable to specific metal-support combinations (e.g., Pt on TiO₂, Fe on N-C), necessitating extensive screening efforts to develop new systems (*Angew. Chem. Int. Ed.* **2024**, e202405017). However, since liquid metals and most metals possess good affinity, our strategy exhibits near-universal applicability. And driven by thermodynamics (the negative mixing entropy), any target metal can be dispersed as single atoms in the liquid metal.

3) Near-100% atomic utilization efficiency for the active metal. Conventional synthesis routes incur active metal loss during high-temperature precursor volatilization and

acid-washing purification. In our method, the dissolution of active metal within the liquid metal suppresses volatilization, while the synthesis itself constitutes a purification process, entirely eliminating acid-washing steps required to remove nanoparticles in traditional methods. Consequently, virtually all introduced active metal atoms convert to utilizable single-atom sites.

4) Highly uniform active site microenvironments enable precisely tunable catalytic performance. On conventional solid supports (e.g., porous carbon, metal oxides), anchoring sites (vacancies, doped N atoms, surface hydroxyl groups) exhibit inherent heterogeneity in size, electronegativity, and spatial geometry. Therefore, the generated single-atom active sites do not have a single structure, and their catalytic activity and selectivity will be weakened, making it difficult to conduct precise structure-activity relationship studies. Conversely, the homogeneous bulk environment of liquid metals ensures a highly uniform coordination geometry for every active-metal atom, thereby guaranteeing high selectivity in subsequent catalytic processes. Liquid metal provides an ideal platform for atomic-level catalytic design and modulation.

Comment 6: *A few specific minor errors: 1) Redefining abbreviations: the authors have already defined the single-atom catalysts and given the abbreviation in line 2 of the abstract, but it is abbreviated again in line 5. 2) Spelling mistake: in Supplementary Fig. 1, the word “luctuates” may be misspelled. 3) Other error: in Supplementary Fig. 11, the x-coordinate is incorrect.*

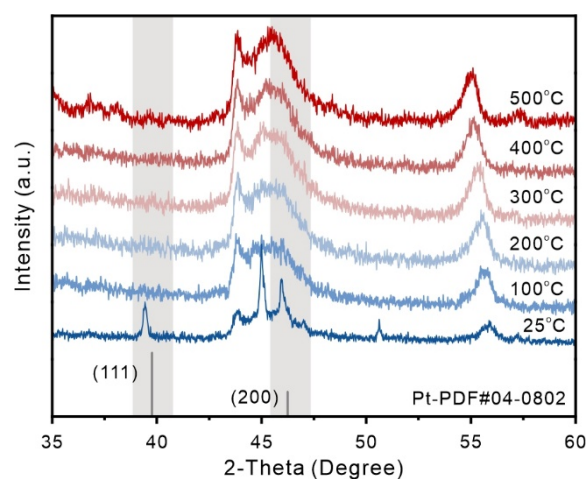
Author reply: Thank you for your kind reminding. We have carefully examined the whole manuscript and corrected the errors to make our manuscript more complete.

Changes: In our revised manuscript, we have addressed all the issues raised by the reviewer, including repeated abbreviations, spelling mistakes, and incorrect axis numbers.

1) Page 2: Unfortunately, SACs are fundamentally constrained by the stability owing to the severe aggregation of single atoms, especially under the high-temperature thermocatalysis operations, which compromises the overall catalytic performance.

2) The caption for Supplementary Fig. 1 on page 6: In the Ga₉₀Pt₁₀ system, the shortest Pt–Pt distance fluctuates between 2.51 Å and 3.65 Å, which are partly in the range of 1.2 times the length of the Pt–Pt bond (2.65 ~ 2.79 Å)⁵.

3) Supplementary Fig. 11 on Page 16:



Supplementary Fig. 11 | *In situ* XRD patterns of Pt@Ga on ZSM-5 in the process of heating up. Initially, the characteristic peaks of Pt can be apparently observed, which are located at 39.8° and 46.3°. With the temperature increasing, the characteristic peaks of Pt were gradually weakened and disappeared subsequently, proving the dispersion of Pt particles.

Finally, we thank the reviewer's suggestion to make our manuscript more rigorous, and we hope these revisions make our manuscript more understandable and informative to readers. Please find our revisions marked in the red copy of the revised manuscript.

To Referee #3

First of all, we really appreciate your constructive comments, which significantly promote the improvement of our work. According to your suggestions, we have made corresponding revisions to our manuscript as listed below.

Overall remark: *This work reports liquid metal dispersed single-atom catalyst by dissolving Pt into molten Ga. In fact, it is the Pt-Ga nanoalloy in previous work. I cannot make sure this kind of nanoalloy with dissolving Pt into molten Ga is the single-atom catalyst. I do not agree with the definition of molten nanoalloy as liquid metal dispersed single-atom catalyst. Thus, I do not think this work should be published in Nature Communications.*

Author reply: Thank you for your kind reminding. The catalyst proposed in our manuscript can be classified as a single-atom alloy (SAA) catalyst, which is a kind of SACs. To clarify the rationale for categorizing our catalyst, we combined the definition of catalysts and their structural characteristics to illustrate. We have carefully checked the references to clarify the definitions of alloy, nanoalloy, single-atom catalysts (SACs), and single-atom alloys (SAAs). Besides, the experimental characterization evidence was also provided to assist in proving the single-atom state of Pt.

1) Firstly, the catalyst proposed in our work conforms to the core definition of the single-atom alloy catalysts that fall within the category of SACs. We systematically analyzed the fundamental distinctions between alloys, nanoalloys, SACs, and SAAs to address reviewers' concerns (Fig. R6). **Alloy catalysts** have a long history and have been developing over nearly 100 years (*J. Chem. Educ.* **1929**, 6, 2097). Building upon the classical definition wherein alloys are most conveniently defined as a metallic system containing two or more components, irrespective of their intimacy of mixing or precise manner of mixing (V. Poncet, G.C. Bond, *Catalysis by Metals and Alloys*, Elsevier, Amsterdam, 1995; *Applied Catalysis A: General* **2001**, 222, 31). When the size of the alloy is reduced to the nanoscale level, they are named “**nanoalloys**” and their properties are distinctly different from the bulk or single metal counterparts (*Adv. Mater.* **2018**, 30, 1705698). Specifically, when dopant metal atoms are isolated completely by the host metal atoms, the alloys are defined as the **SAAs** (*Science* **2012**, 335, 1209; *Phys. Chem. Chem. Phys.* **2013**, 15, 12187; *Chem. Rev.* **2023**, 123, 5859). In addition, SAAs also constitute a subset of **SACs** (*Mater. Today Cataly.* **2023**, 2, 100009).

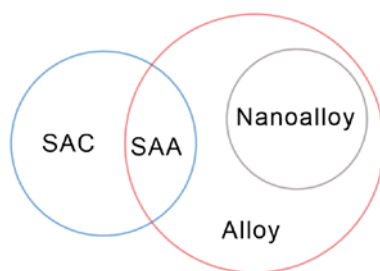


Fig. R6 | A summary of the definitions of alloy, nanoalloy, SAC and SAA.

2) Secondly, the monodispersion state of Pt atoms within Ga has been confirmed by both X-ray absorption fine structure (XAFS) measurements and theoretical simulations, which strictly complies with the core definition of SACs. XAFS results have clearly demonstrated that the Pt–Pt bond is absent and there exists the Ga–Pt bond in the liquid metal-based catalyst (Fig. R7), indicating that Pt atoms exist in the form of single atoms. Meanwhile, through ab initio molecular dynamics (AIMD) simulations under high-temperature conditions, it is shown that the average distance between Pt atoms and Ga atoms is much longer than the intrinsic bond length of Pt–Pt (2.65–2.79 Å) (Fig. R8), and no formation of Pt dimers or clusters has been detected in the dynamic statistics.

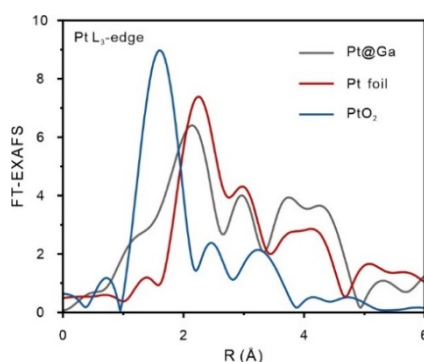


Fig. R7 | FT Pt L_3 -edge EXAFS spectra in R space of liquid catalyst with phase correction.

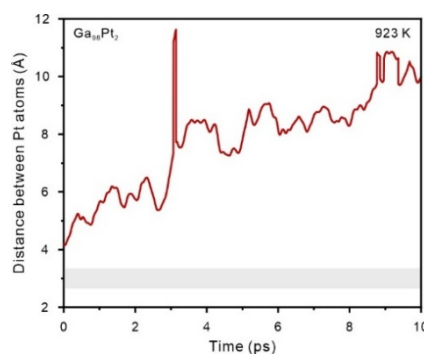


Fig. R8 | Time evolution of Pt–Pt interatomic distance in the $\text{Ga}_{98}\text{Pt}_2$ system under NVT ensemble simulation.

Changes: Thanks for your comment. We have acknowledged the previous research on liquid metals conducted by others and have included references to it in the text. For your reference, please find our revisions marked in the red copy. Once again, we apologize for our lack of rigor.

Comment 1: *I suggest the authors to make sure the definition of single-atom catalyst. This kind of molten Pt–Ga nanoalloy is not new, and their catalytic mechanism is the contribution of both Pt and Ga. It is NOT the case of single-atom catalyst.*

Author reply: Thank you for your kind reminding. Based on the definition of SACs and supported by our experimental results, our catalyst is the case of SACs.

1) Firstly, **the structure of Pt–Ga catalyst in our work conforms to the core definition of SACs.** The concept of “SACs” was first proposed by Zhang, Li, Liu, and co-workers in 2011 (*Nat. Chem.* **2011**, 3, 634). The SACs, as defined originally, are “one kind of supported metal catalyst with isolated metal atom sites, where the core metal atoms in the active sites are not closely connected but interact with support” (*Acc. Chem. Res.* **2025**, 58, 1878). In our manuscript, the **experimental results all confirm the isolated nature of the Pt atoms in our catalyst (Fig. R6 and Fig. R7), which is consistent with the core definition of SACs.**

2) Secondly, **the catalytic performance of this Pt–Ga catalyst is mainly contributed by Pt atoms.** SACs predominantly rely on monodispersed atomic sites as active centers, where the support primarily serves to anchor single atoms and modulates the electronic states of isolated atoms through interfacial charge transfer effects. In our manuscript, liquid metal not only disperses Pt single atoms, but its dynamic fluidity can also promote the movement of single atoms. The **activity of the catalyst mainly stems from the isolated Pt single-atom sites dispersed in Ga.** The liquid metal was introduced to achieve atomic dispersion of the active metal, establishing a highly uniform coordination environment of Pt to ensure superior catalytic selectivity. Meanwhile, the adoption of porous supports is mainly to enhance the gas mass transfer efficiency, which we have already clarified in our manuscript. Porous supports are universally employed during catalytic processes, as they guarantee a relatively chemical inertness and the absence of mass-transfer restrictions because of the large pore size and rich pore structure (*Nat. Chem.* **2017**, 9, 862). By comparing the conversion and selectivity of pure ZSM-5, pure Ga, and Pt@Ga/ZSM-5, it is obvious that both ZSM-5 and Ga have no impact on the activity. The main catalytic active center is still Pt (Fig. R9).

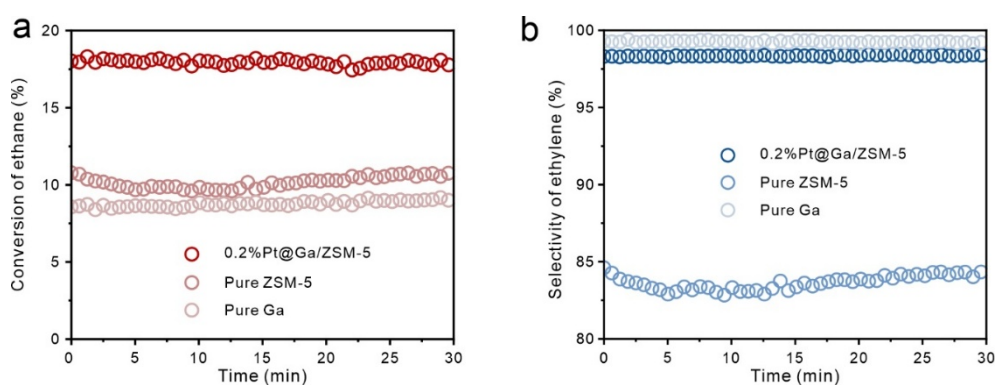


Fig. R9 | Ethane dehydrogenation performance of liquid metal dispersed SACs. a The conversion of ethane. **b** The selectivity of ethylene.

Hence, the Pt single atoms dispersed by liquid Ga conform to the core definition of SACs (atomic-level dispersion, isolated active sites), can be considered a kind of dynamic SACs.

Changes: Thanks for your comment. We have acknowledged the previous research on liquid metals conducted by others and have included references to it in the text. For your reference, please find our revisions marked in the red copy.

Comment 2: *The mechanism of high temperature stability is not clear. The authors need to prepare a single Pt atom in molten Ga, and Pt almost remain the top of the molten Ga cluster and also high temperature stability.*

Author Reply: Thank you for your valuable suggestion. Following your suggestion, the stability mechanism of Pt single atom on Ga was demonstrated through the theoretical calculations. We constructed the simulation model, in which the atom ratio and temperature are strictly consistent with those in the experiment. The results confirmed the single-atom dispersion state of Pt atoms in liquid metal, consistent with the experimental findings.

1) Theoretical calculations have proved that a single Pt atom in molten Ga can freely migrate in the molten Ga, and there is a certain probability for it being exposed on the Ga surface. To validate this mechanism, we constructed a $\text{Ga}_{79}\text{Pt}_1$ slab model. AIMD simulations were conducted to investigate the motion trajectory of a single Pt atom in molten Ga (analyzing its vertical distance to the surface). Throughout the 10 ps simulation period, a single Pt atom migrates relatively freely within molten Ga, with a depth from the Ga surface ranging from 0 to 4.84 Å (Fig. R7). Statistical analysis indicates a surface exposure probability of 56.3% for the isolated Pt atom. Furthermore, considering the susceptibility of metallic Ga to oxidation state, even under reducing atmospheres (e.g., H_2) at 923 K, a thin

Ga₂O₃ layer may persist. To eliminate the influence of Ga oxidation on Pt exposure, we also constructed a Ga₇₉Pt₁O₁₀ slab model incorporating a small number of oxygen atoms and performed a 10 ps AIMD simulation. The simulation results demonstrate that limited oxygen atoms form Ga₂O₃ clusters with surface Ga atoms. Under the influence of these Ga₂O₃ clusters at the molten Ga surface, the Pt atom primarily remains within the surface and subsurface regions, exhibiting a depth range of 0 to 2.97 Å (Fig. R10). Notably, the surface exposure probability of the single Pt atom increases significantly to 81.3%. The single Pt atom in liquid metal migrates freely as Ga₈Pt or Ga₉Pt clusters, retaining a finite probability of surface exposure. Crucially, the presence of Ga₂O₃ clusters at the molten Ga surface dramatically enhances the probability of single Pt atom surface exposure. This indicates that the **Pt single atom can be stabilized to the molten Ga surface at 923 K over extended periods, thereby serving as active sites for the catalytic activation of C₂H₆ molecules.**

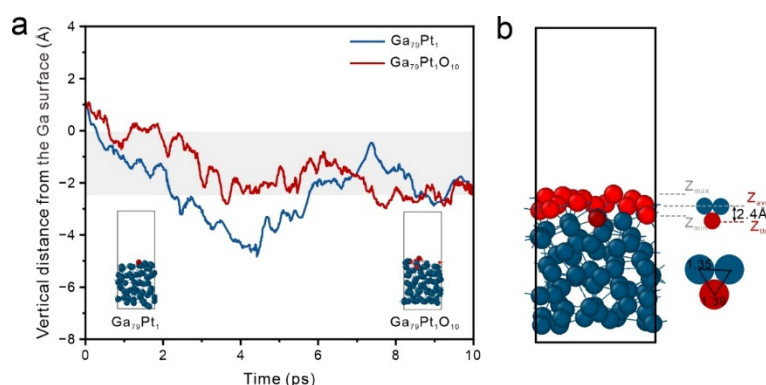


Fig. R10 | The distance between a single Pt atom and the surface Ga layer. **a** The calculated curve of the distance between a single Pt atom and the surface Ga layer as a function of time. **b** The illustration of the distance between a single Pt atom and the surface Ga layer. Here, surface Ga atoms are characterized as top-layer Ga atoms in each frame structure whose coordination number is two less than the maximum coordination number, with the zero point in the z-direction defined as the average of the maximum and minimum z-coordinates of these surface Ga atoms. Furthermore, the z-coordinate threshold for identifying exposed Pt atoms is defined as the vertical distance corresponding to the closest packing radius of Re atoms relative to this zero reference point.

2) The mechanism of high temperature stability stems from the stable interaction between Pt and Ga, which is validated by both thermodynamics and dynamics. Firstly, thermodynamic analysis reveals that the system exhibits a lower total energy for Pt monodispersed in liquid metal than as clusters (Fig. R11a). Therefore, the remarkable atomic diffusion of liquid Ga metal at high temperature promotes the migration of heterogeneous active metal atoms, thereby stabilizing their dispersed atomic state in liquid

Ga rather than forming dimers and trimers inside the molten Ga. Secondly, dynamic simulations of Pt atoms diffusion within Ga demonstrate that Pt atoms are resistant to aggregation. The Pt–Pt interatomic distance progressively increases from 4.18 Å to 11.63 Å (Fig. R11b), significantly exceeding the typical Pt–Pt bond length range of 2.65–2.79 Å (*J. Am. Chem. Soc.* **1978**, *100*, 2074). All these results confirm that Pt atoms maintain stable single-atom states under high temperature.

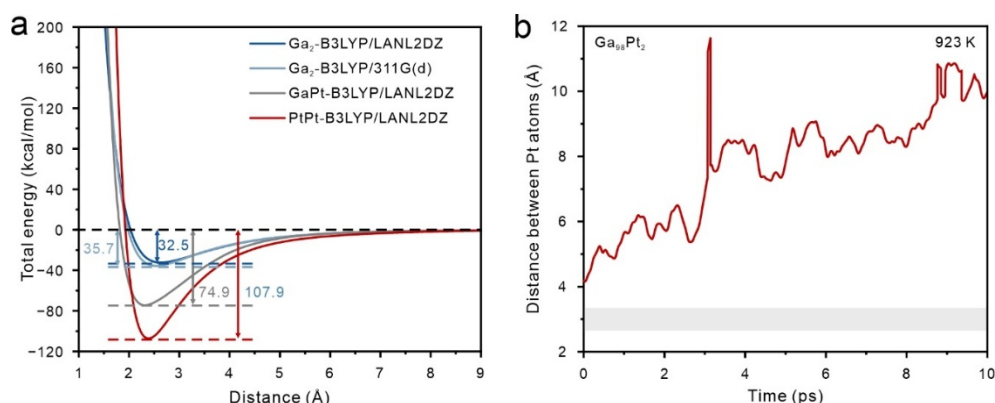


Fig. R11 | The thermodynamics and kinetics simulations of Pt atoms diffusion within Ga. **a** Total energy versus time curves for the $\text{Ga}_{90}\text{Pt}_{10}$ system containing clustered Pt_{10} and dispersed Pt_{10} . **b** Time evolution of Pt–Pt interatomic distance in the $\text{Ga}_{98}\text{Pt}_2$ system under NVT ensemble simulation.

3) Stability evaluation of liquid metal dispersed SACs was evaluated in non-oxidative ethane dehydrogenation at 923 K. It exhibits sustainable conversion and selectivity in the continuous reaction for 100 h, where the constant ethane conversion of ~18% is obtained without attenuation (Fig. R12a). Furthermore, the EXAFS of liquid metal dispersed SACs after 100 h confirmed the preservation of atomically dispersed Pt within Ga (Fig. R12b, c). Hence, comprehensive stability evaluations combined with post-reaction structural analysis reveal that Pt maintains atomic dispersion while participating in the catalytic process.

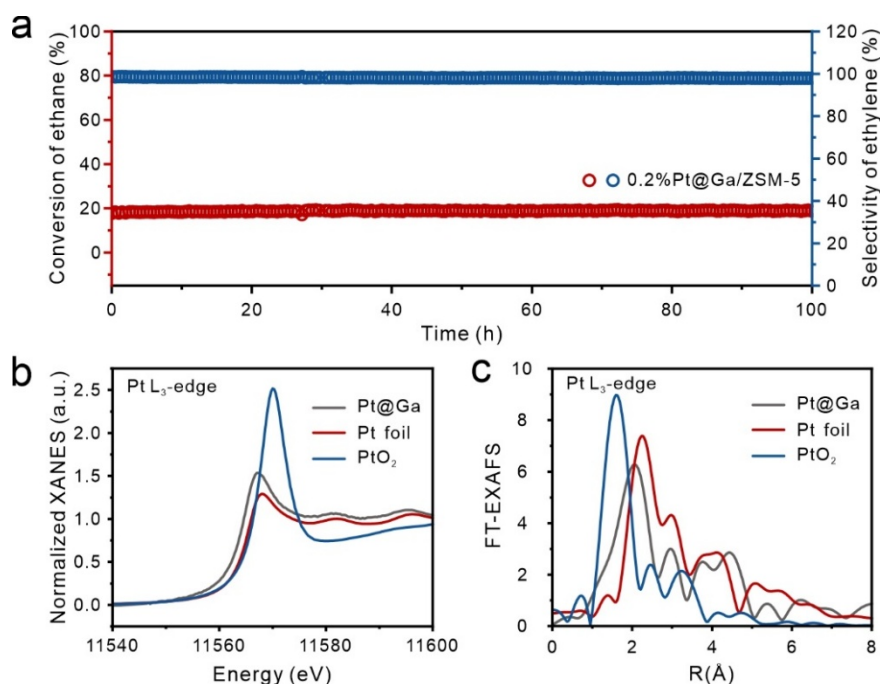


Fig. R12 | Thermal stability performance of liquid metal dispersed SACs. **a** The long-term stability test in ethane dehydrogenation. **b** XANES spectra of the liquid metal dispersed SACs after 100 h ethane dehydrogenation taken at Pt L_3 edge. **c** k_3 -weighted FFT spectra in R -space at Pt L_3 edge.

Comment 3: *The data analysis for liquid metal dispersed single-atom catalyst is not enough for understanding high temperature stability. How to certify that the Pt atoms do not form dimer and trimer inside the molten Ga after long-time high temperature test?*

Author reply: Thank you for your constructive comment. As you put forward, we have used both theoretical calculations and detailed experimental structural characterization to prove that the Pt atoms don't form dimers or trimers inside the molten Ga after long-time high temperature test, hoping to resolve the reviewer's doubt as much as possible.

1) Thermodynamic analysis reveals that the system exhibits a lower total energy for Pt monodispersed in liquid metal than as clusters. Using Pt_{10} as a model of Pt cluster, we investigated the structural evolution of both clustered and dispersed Pt_{10} in liquid Ga during 10 ps simulations at 923 K, in which the temperature is consistent with the reaction temperature of ethane dehydrogenation in experiment. And calculated total energy versus time curves for the $Ga_{90}Pt_{10}$ system containing clustered Pt_{10} and dispersed Pt_{10} (Fig. R13). The dispersed Pt_{10} generally exhibits lower total energy compared to the aggregated Pt_{10} clusters within the simulated 10 ps. The results indicate that the remarkable atomic diffusion of liquid Ga metal at high temperature promotes the migration of heterogeneous active metal atoms, thereby stabilizing their dispersed atomic state in liquid Ga rather than forming dimers and trimers inside the molten Ga.

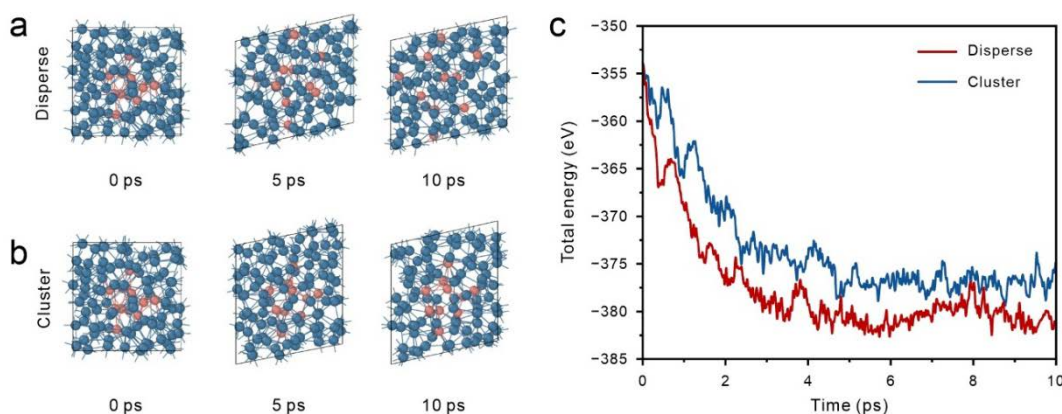


Fig. R13 | Theoretical calculations of heterogeneous active metal atoms monodispersed in liquid Ga. **a, b** The extracted 0 ps, 5 ps, and 10 ps snapshots from AIMD trajectories of both clustered (**a**) and dispersed (**b**) Pt_{10} in liquid Ga. **c** Total energy versus time curves for the $\text{Ga}_{90}\text{Pt}_{10}$ system containing clustered Pt_{10} and dispersed Pt_{10} .

2) Dynamical simulations of Pt atoms diffusion within Ga demonstrate that Pt atoms are resistant to aggregation. The actual existence state of Pt atoms in liquid Ga was studied using AIMD simulations on the $\text{Ga}_{98}\text{Pt}_2$ system (Pt dimer embedded in Ga_{98}) at 923 K. The 10 ps NVT ensemble simulations (Fig. R14) demonstrate that the Pt–Pt interatomic distance progressively increases from 4.18 Å to 11.63 Å, significantly exceeding the typical Pt–Pt bond length range of 2.65–2.79 Å (*J. Am. Chem. Soc.* **1978**, 100, 2074). These results confirm that Pt atoms maintain stable single-atom states under high temperature.

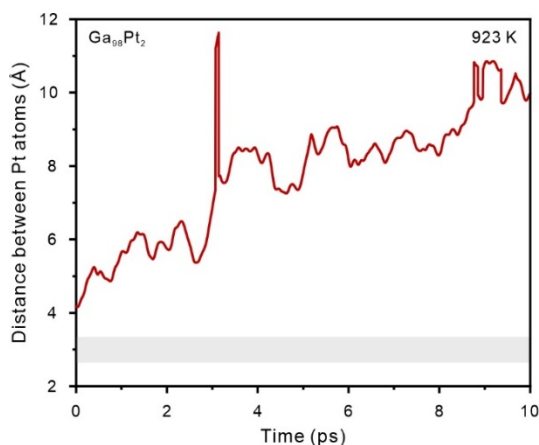


Fig. R14 | Time evolution of Pt–Pt interatomic distance in the $\text{Ga}_{98}\text{Pt}_2$ system under NVT ensemble simulation.

3) The exact dispersion state of Pt in the Pt@Ga catalyst was further verified by X-ray absorption spectroscopy (XAS) measurements. The k_3 -weighted R -space Fourier transformed spectra of the extended X-ray absorption fine structure spectra (EXAFS) show the prominent peak at 2.14 Å for Pt@Ga SACs, distinguished from that of the Pt–O (1.62

Å) bond for PtO₂ and the Pt–Pt bond (2.24 Å) for Pt foil (Fig. R15a, b). Notably, the XANES spectrum and *R*-space of liquid metal dispersed SACs after 100 h ethane dehydrogenation at Pt *L*₃-edge are similar to that of pristine liquid SACs (Fig. R15c, d), demonstrating exceptional structural integrity under prolonged thermal catalysis. The absence of Pt–Pt bonds definitively confirms the exclusion of dimers or trimers.

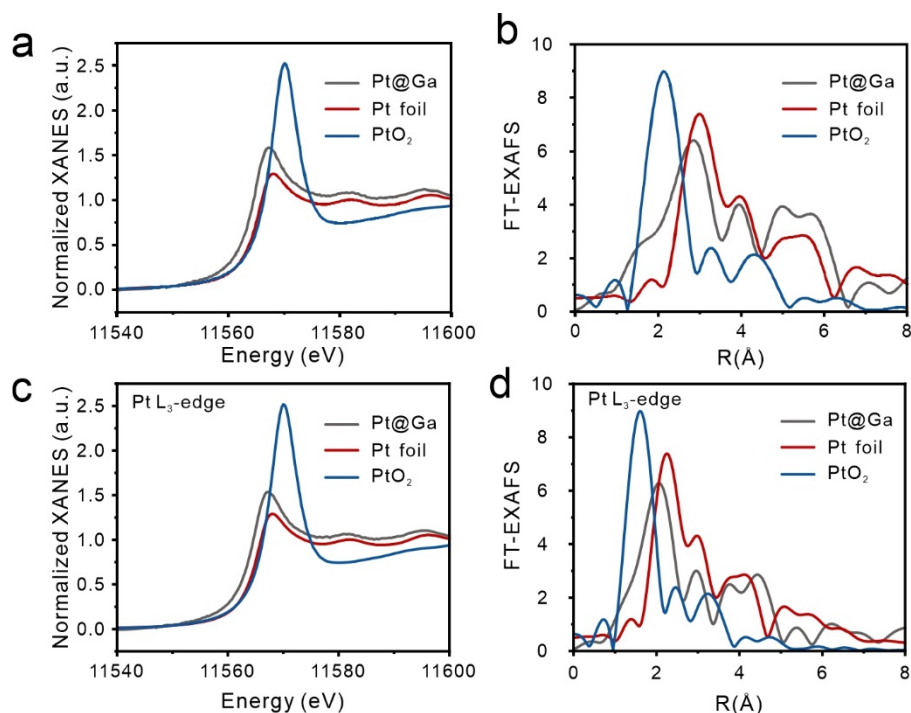


Fig. R15 | Normalized XANES spectra at the Pt *L*₃-edge and FT Pt *L*₃-edge EXAFS spectra in *R* space of liquid catalyst before and after the ethane dehydrogenation.

4) The long-term stability test of liquid metal dispersed SACs further verified the stability of the atomically dispersed configuration. During the catalytic process, the state of the active center directly affects the catalytic pathway. Usually, the formation of Pt particles leads to deep dehydrogenation of alkanes, resulting in a decrease in the selectivity and conversion of the catalyst. In our work, the performance of the liquid metal dispersed SACs in the durability test exhibits sustainable conversion and selectivity in the continuous reaction for 100 h, where the constant ethane conversion of ~18% is obtained without attenuation (Fig. R16). These results provide indirect evidence that the catalyst maintains a stable single-atom state.

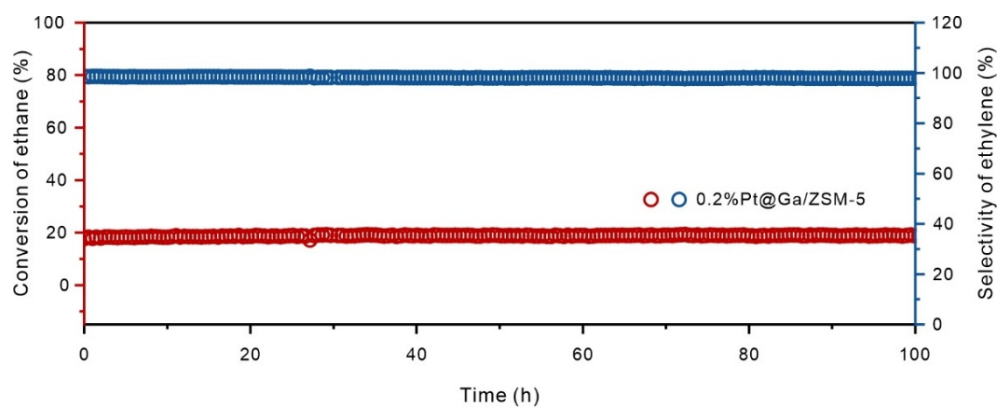


Fig. R16 | Long-term stability test of liquid metal dispersed SACs.

Finally, we want to thank the referees again for their constructive comments on our work, and we hope our revised manuscript can reach the quality expectation to be published in *Nature Communications*. Please find our revisions marked in the red copy of the revised manuscript.

Reply to the referees

To Referee #2

First of all, we really appreciate your high evaluation to recommend our work to be published in *Nature Communications*.

Overall remark: *I read the manuscript carefully and found that the authors did a good job of revising it. The new version looks much better compared to the original one. As my concerns were fully addressed in this revision, I believe the manuscript is now publishable.*

Author reply: Thank you very much for your positive feedback on our revised manuscript. We are delighted to hear that you found the new version to be a significant improvement and that your previous concerns have been fully addressed. We appreciate your recommendation for publication.

To Referee #3

We have made a point-by-point response to your comments and carefully revised the manuscript as you suggested. For your reference, please find our revisions marked in red.

Overall remark: *I agree with Reviewer #1 that a single Pt atom can be stabilised in liquid gallium to perform high-temperature dehydrogenation catalysis is not new. The revision still lacks the required novelty for a publication in Nature Communications. I also agree with Reviewer #3 that the definition of molten nanoalloy is not liquid metal dispersed single-atom catalyst. This kind of molten Pt–Ga nanoalloy is not new, and should not be published in Nature Communications.*

Author reply: It is possible that the core innovation of our work may have been inadvertently understated. Our work focuses on **the severe sintering limitation of single-atom catalysts (SACs) at high temperatures in industrial catalysis. By leveraging the fluidity and metal affinity of liquid metal, we realized the ultra-stable SACs at 650 °C.** Unlike the reported works that single atoms generally tend to aggregate at high temperatures owing to Ostwald ripening or atom migration (*J. Stat. Phys.* **1985**, 38, 231; *Acc. Chem. Res.* **2013**, 46, 1720; *Nat. Catal.* **2021**, 4, 469; *Nat. Catal.* **2025**, 8, 1038), it is worth noting that **the high temperature contributes to the dispersion single atoms profited from the affinity between Ga and Pt in our work.** Besides, the **Pt–Ga catalyst in our work is exactly a kind of SACs.** Regarding your question, we have made a point-by-point response, hoping to clear up your doubts.

1) As for the definition of molten nanoalloy is not liquid metal dispersed SAC, we would like to clarify. The catalyst proposed in our work aligns with the established definition of single-atom alloy catalysts (SAAs), which is a kind of SACs. And this has been evidenced by both theoretical calculations and experiments.

The Pt–Ga catalyst in our work conforms to the core definition of SACs. The concept of “SACs” was first proposed by Zhang, Li, Liu, and co-workers in 2011, in which isolated single atoms were anchored onto nano-crystallites (*Nat. Chem.* **2011**, 3, 634). Up to now, the SACs are universally regarded as “one kind of supported metal catalyst with isolated metal atom sites, where the core metal atoms in the active sites are not closely connected but interact with support” (*Acc. Chem. Res.* **2025**, 58, 1878). The concept of “SAAs” was first put forward in 2012 by Georgios Kyriakou et al. with two key characteristics, which fall within the category of SACs (*Science* **2012**, 335, 1209).

“Two key characteristics of SAAs are (i) the more active of the two components is present in the surface of the host metal at very low concentrations [~ 0.01 monolayer (ML)] and (ii) atoms of the more active component are thermodynamically more stable when surrounded by the host metal such that no dimers or trimers are present at low coverage.” Currently, it is commonly accepted that SAAs are a class of single-site catalysts in which small amounts of isolated metal atoms are present in the surface layer of a metal host. Typically, they are comprised of single atoms of a catalytically active metal alloyed into the surface of a less reactive host metal (*Chem. Rev.* **2020**, *120*, 12044). Based on these criteria, we conducted a rigorous comparison between the catalyst in our work and the definition of SAAs, confirming that the catalyst in our work fully satisfies the requirements: (a) **The Pt atoms are presented as the single-atom state within Ga and dynamically migrate onto the surface to participate in the catalytic reaction** (which was proved theoretically and in experiment); and (b) **the Pt atoms are mainly active centers** (which was evidenced by the experiments). The detailed analysis is as follows:

i) Thermodynamic analysis reveals that Pt is more likely to exhibit as the single-atom dispersion state within Ga from the perspective of thermodynamics. The dispersed Pt atoms generally exhibit lower total energy compared to the aggregated Pt clusters within the simulated 10 ps (Fig. R1).

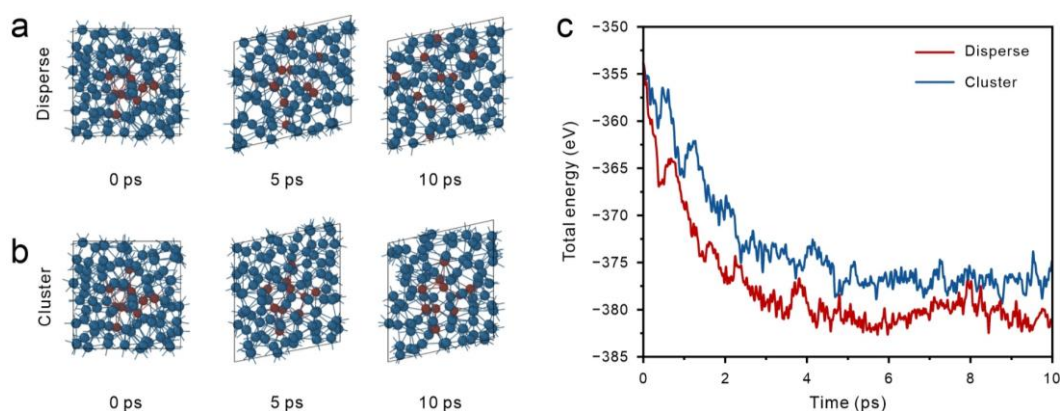


Fig. R1 | Theoretical calculations of heterogeneous active metal atoms monodispersed in liquid Ga. **a, b** The extracted 0 ps, 5 ps, and 10 ps snapshots from AIMD trajectories of both clustered (**a**) and dispersed (**b**) Pt_{10} in liquid Ga. **c** Total energy versus time curves for the $\text{Ga}_{90}\text{Pt}_{10}$ system containing clustered Pt_{10} and dispersed Pt_{10} .

ii) **Dynamical simulations of Pt atom diffusion within Ga demonstrate that Pt atoms are resistant to aggregation.** The actual existence state of Pt atoms in liquid Ga was studied using AIMD simulations on the $\text{Ga}_{98}\text{Pt}_2$ system (Pt dimer embedded in Ga_{98}) at 923 K. The 10 ps NVT ensemble simulations (Fig. R2) demonstrate that the Pt–Pt interatomic distance progressively increases from 4.18 Å to 11.63 Å, significantly exceeding the typical Pt–Pt bond length range of 2.65–2.79 Å (*J. Am. Chem. Soc.* **1978**, *100*, 2074). These results confirm that Pt atoms maintain stable single-atom states under high temperature.

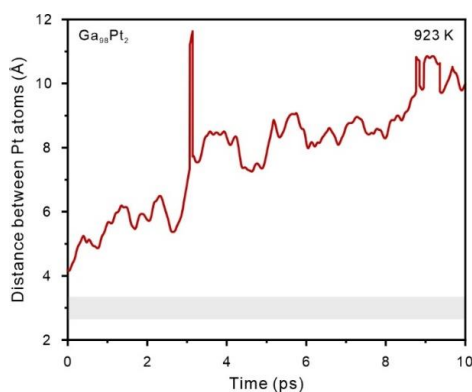


Fig. R2 | Time evolution of Pt–Pt interatomic distance in the $\text{Ga}_{98}\text{Pt}_2$ system under NVT ensemble simulation.

iv) The single-atom dispersed state of Pt atoms in the Pt–Ga catalyst was proved by X-ray absorption fine structure (XAFS). XAFS results have clearly demonstrated that the Pt–Pt bond is absent and there only exists the Ga–Pt bond in the liquid metal-based catalyst (Fig. R5), indicating that Pt atoms exist in the form of single atoms.

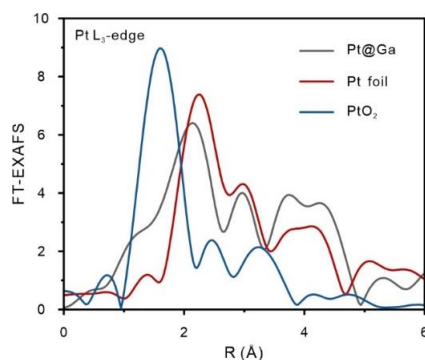


Fig. R5 | FT Pt L_3 -edge EXAFS spectra in R space of liquid catalyst with phase correction.

Secondly, the Pt atoms are mainly active centers. SACs predominantly rely on monodispersed atomic sites as active centers, where the support primarily serves to anchor single atoms and modulates the electronic states of isolated atoms through interfacial charge transfer effects. In our manuscript, liquid metal not only disperses Pt single atoms, but its dynamic fluidity can also promote the movement of single atoms. **The activity of the catalyst mainly stems from the isolated Pt single-atom sites dispersed in Ga.** The liquid metal was introduced to achieve atomic dispersion of the active metal, establishing a highly uniform coordination environment of Pt to ensure superior catalytic selectivity. Meanwhile, the adoption of porous supports is mainly to enhance the gas mass transfer efficiency, which we have already clarified in our manuscript. Porous supports are universally employed during catalytic processes, as they guarantee a relatively chemical inertness and the absence of mass-transfer restrictions because of the large pore size and rich pore structure (*Nat. Chem.* **2017**, 9, 862). By comparing the conversion and selectivity of pure ZSM-5, pure Ga, and Pt@Ga/ZSM-5, it is obvious that both ZSM-5 and Ga have no impact on the activity. The main catalytically active center is still Pt (Fig. R6).

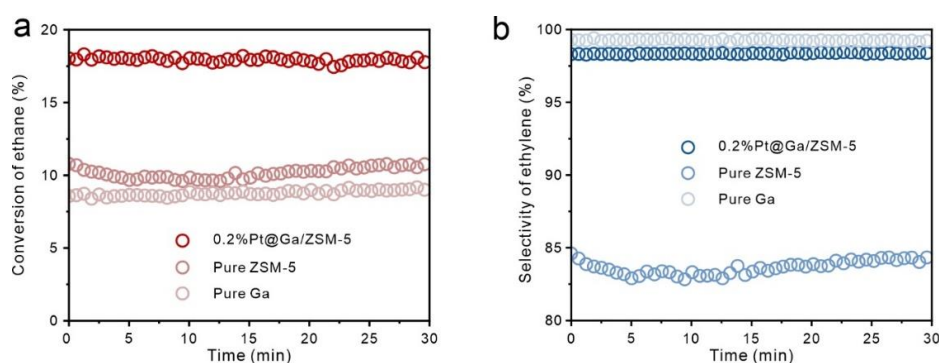


Fig. R6 | Ethane dehydrogenation performance of liquid metal dispersed SACs. a The conversion of ethane. **b** The selectivity of ethylene.

Therefore, **the detailed structural characterization and theoretical calculation results have both proved that this catalyst conforms to the core definition of SAAs—a kind of SACs.**

2) In terms of the query that a single Pt atom can be stabilized in liquid Ga, the reviewer considered that it has been reported. We would like to clarify.

Our work focuses on the severe sintering limitation of SACs at high temperatures in industrial catalysis. By leveraging the fluidity and metal affinity of liquid metal, we realized the ultra-stable SACs at 650 °C. It is worth noting that the high temperature contributes to the dispersion of single atoms in our work rather than the agglomeration sintering phenomenon in reported works, which is of great significance for promoting the industrial application of SACs. The significance of our work lies as follows:

i) The realization of highly thermal-stable SACs for industrial catalysis. SACs have emerged as promising candidates to replace conventional industrial catalysts owing to their exceptional catalytic activity and atomic efficiency. The high surface energy of single atoms renders them prone to migration and subsequent sintering during catalytic processes, especially at high temperatures. Thus, it remains **a formidable obstacle to developing novel catalysts with high-temperature thermal stability.** We realized the highly thermal-stable SACs resistance to sintering at harsh conditions through harnessing the inherent metal affinity and fluidity of liquid metal (Fig. R7). We believe this is a crucial step forward in promoting the application of SACs in high-temperature industrial catalysis.

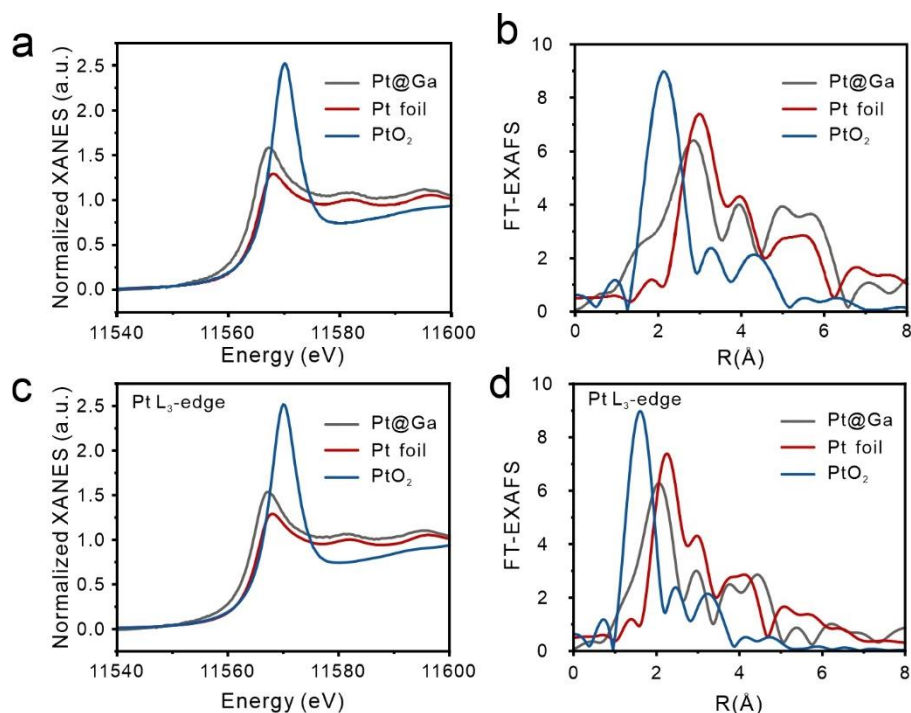
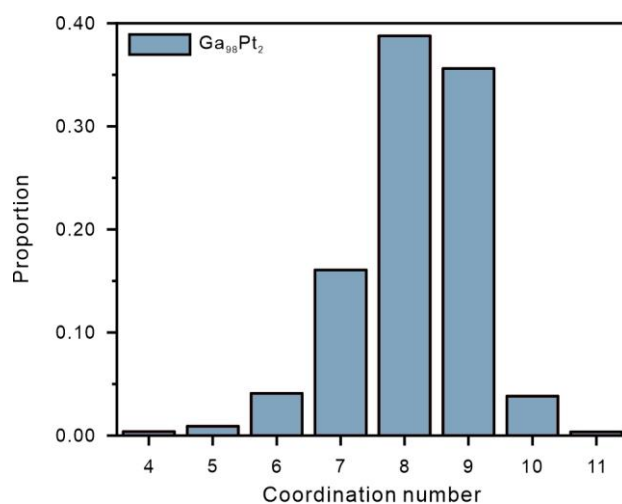


Fig. R7 | Normalized XANES spectra at the Pt L_3 -edge and FT Pt L_3 -edge EXAFS spectra in R space of liquid catalyst before and after the ethane dehydrogenation.

ii) A brand-new strategy for achieving highly thermal-stable SACs resistance to sintering by liquid metals. The mono-dispersion of Pt in liquid metals rather than agglomeration at high temperatures is a thermodynamically favourable process, which is owing to the selective breaking and formation of chemical bonds profiting from the inherent metal affinity of liquid metal. Combined with the fluidity of liquid metal, active metal atoms can move but remain confined to the liquid metal as the metallic single-atom state at high temperatures. We have provided robust evidence for clarifying the mechanism (Figure 2 and Supplementary Mov. S1 and S2 in our revised manuscript). Besides, X-ray absorption fine structure (XAFS) analysis provides unequivocal proof to verify the single-atom state of Pt in Ga before and after the ethane dehydrogenation (Fig. R5).

Regarding the doubts about the reported works, we have conducted a comprehensive literature search, retrieving studies on liquid metal Ga and Pt for dehydrogenation catalysis. Previous reports focus on the catalytic performance of the Pt in liquid metal limited in low-temperature or the catalytic performance of the alloy state in thermal catalysis (*Nat. Chem.* **2022**, *14*, 935; *ACS Appl. Nano Mater.* **2024**, *7*, 24356; *Fuel*, **2024**, *369*, 131788; *ACS Catal.* **2021**, *11*, 13423). It is finding that our work first proposed and confirmed that liquid metals can solve the problem of severe sintering of

single atoms under extremely high-temperature conditions, which was not reported before. Therefore, the **core innovation of work and the previously reported works are essentially different**. We don't think these articles conflict.



Supplementary Fig. 4 | Statistical distribution of Pt coordination numbers in $\text{Ga}_{98}\text{Pt}_2$ from AIMD simulations. The coordination environment of Pt atoms in the $\text{Ga}_{98}\text{Pt}_2$ system shows that the dominant coordination numbers for individual Pt atoms are 8 or 9, implying that mono-dispersed Pt atoms migrate within liquid Ga in the form of clusters such as Ga_8Pt or Ga_9Pt .

Reply to the referees

To Referee #4

First of all, thank you very much for your evaluation of our work. According to your suggestion, we have supplemented a point-to-point reply with systematic experiments and discussions. We hope that our revised manuscript is considered to be published in *Nature Communications*.

Overall remark: *In the manuscript entitled “Liquid metal dispersed single-atom catalyst with high-temperature stability” submitted to Nat. Commun., Zeng et al. report a Pt-based single-atom catalyst dispersed in a Ga-based molten phase and supported on ZSM-5 for ethane dehydrogenation to ethylene. The authors imply that this constitutes a new conceptual way of stabilizing SACs with high activity and thermal stability, and they support their arguments by a combination of experimental data and AIMD calculations.*

Author reply: Thanks a lot for your evaluation that “this constitutes a new conceptual way of stabilizing SACs with high activity and thermal stability, and they support their arguments by a combination of experimental data and AIMD calculations.” Regarding the following detailed questions that you put forward, we have supplemented systematic experiments and discussions, including the core innovation of our work, the generalizability of our findings, reaction kinetic analysis, the long-term stability, deactivation pathways, and structure-performance relationships. We believe that the novelty, generalizability, and depth of our work have been greatly improved by your insightful suggestion.

Comment 1: *My main concern is that the central experimental claim about the unique role of Pt is not convincingly supported by the data as described. The authors state in their rebuttal letter that, upon “comparing the conversion and selectivity of pure ZSM-5, pure Ga, and Pt@Ga/ZSM-5, it is obvious that both ZSM-5 and Ga have no impact on the activity.” This is difficult to reconcile with the reported performance: Ga metal alone reaches approximately half the conversion of the Pt–Ga single-atom system and even exhibits higher selectivity to ethylene. These numbers strongly suggest that Ga is an active and selective catalyst on its own and that the incremental contribution of Pt to the overall performance is limited. Rather than demonstrating a clear, dominant role of Pt single atoms in a molten Ga host, the data point to a system in which Ga already performs well and Pt provides a modest, enhancement in conversion at (perhaps) the cost of selectivity. The authors need to acknowledge and analyze this more carefully; in its current form, the narrative that Ga and ZSM-5 “have no impact on the activity” is not credible.*

Author reply: Thank you very much for your constructive feedback on our revised manuscript. Regarding your questions about the contribution of Ga, and Pt to the overall ethane dehydrogenation (EDH) performance, we conducted supplementary experiments. The results indicate that Ga and ZSM-5 do not provide additional catalytically active sites for EDH. Ga mainly serves as a dispersion carrier, enabling high catalytic activity by dispersing metal Pt particles into single atoms. And the ZSM-5 was used to enhance the gas mass transfer. The detailed analysis is listed below:

(1) **Regarding whether Ga or ZSM-5 is an active and selective catalyst on its own**, we supplemented experiments with blank U-tubes. Due to the fluidity of liquid metal Ga, conventional U-tubes cannot meet the requirements for EDH testing. In this work, we customized a home-made U-tube equipped with a quartz sieve plate to support the catalyst as well as to enhance the mass transfer (Fig. R1). Under identical catalytic conditions, the catalytic performance of the blank U-tube and pure Ga (pure Ga@U-tube) is shown in Fig. R2. The ethane conversion and ethylene selectivity of the U-tube are derived from the thermal cracking reaction of ethane. Comparison of catalytic performance reveals that the ethane conversion of pure Ga, and the blank U-tube remains essentially consistent, suggesting that Ga does not provide additional catalytically active sites. Regarding the activity ZSM-5, the ZSM-5 (Si/Al = 45) used in our work is usually considered to exhibit negligible activity for ethane dehydrogenation to ethylene, and substantial catalytic performance is typically achieved through constructing tailored active sites via the surface immobilization of metal nanoparticles (*ACS Catal.* **2021**, *11*, 9970) or the encapsulation of clusters or isolated metal sites within its framework (*Appl. Surf. Sci.* **2021**, *554*, 149611; *Chem* **2023**, *9*, 637).

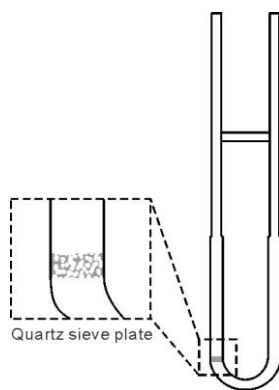


Fig. R1 | Schematic diagram of a quartz tube with a quartz sieve plate. The porous quartz sieve plate is designed to prevent liquid metal leakage.

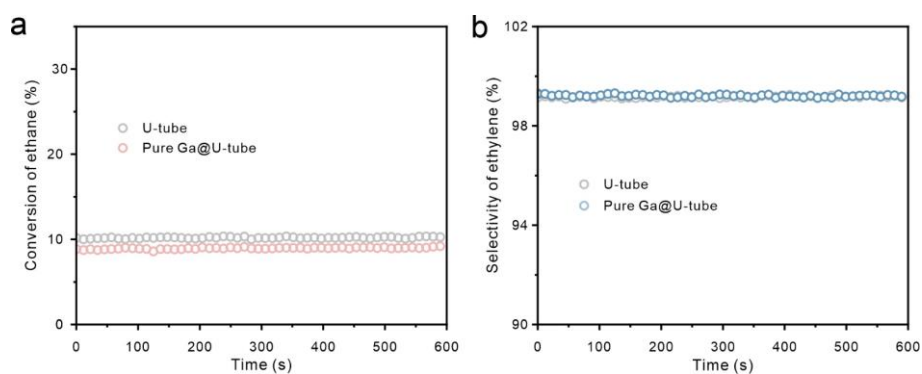


Fig. R2 | Comparison of catalytic performance of blank U-tube, and pure Ga. (a) Conversion of ethane. (b) Selectivity of ethylene.

(2) Ga mainly acts as a dispersion support to enable the monodispersed state of Pt atoms, which enhances the ethane conversion and ethylene selectivity. To further demonstrate the role of Ga, we compared the catalytic performance of Pt particles@ZSM-5 (direct mixing of commercialized Pt particles with ZSM-5 without utilizing Ga as dispersion) with that of Pt SACs@Ga/ZSM-5. Pt particles@ZSM-5 exhibited ethane conversion at 9.5% and ethylene selectivity at 82.6% (Fig. R3). In contrast, the Pt SACs@Ga/ZSM-5 catalyst, where Pt dissolves as single atoms within liquid Ga, achieves a significantly higher ethane conversion (17.8%) while maintaining excellent ethylene selectivity (98.3%). The breakthrough in performance stems from the introduction of liquid metal, which directly alters the dispersed state of Pt atoms (Fig. 2e, f in the revised manuscript). By dispersing Pt particles into single Pt atoms, the utilization of Pt atoms was maximized to achieve high ethane conversion. Furthermore, the coordination environment of active metals significantly influences the reaction selectivity. Usually, Pt particles with a non-uniform coordination environment typically promote C–C cleavage and deep dehydrogenation side reactions (*ACS Catal.* **2020**, *10*, 12932). In contrast, Pt-based SACs with a uniform coordination environment exhibit excellent selectivity for ethylene due to their superior activation of C–H bonds in alkanes, while simultaneously limiting C–C bond cleavage to prevent dehydrogenation reactions. Therefore, the introduction of liquid metal, which stabilizes Pt in a single-atom state with a highly uniform coordination environment, allows the Pt SACs@Ga/ZSM-5 catalyst to reach an ethylene selectivity of 98.3%. Taken together with the evidence that metallic Ga is catalytically inert (Fig. R2), we conclude that Ga acts as a liquid dispersion to scatter and stabilize Pt atoms.

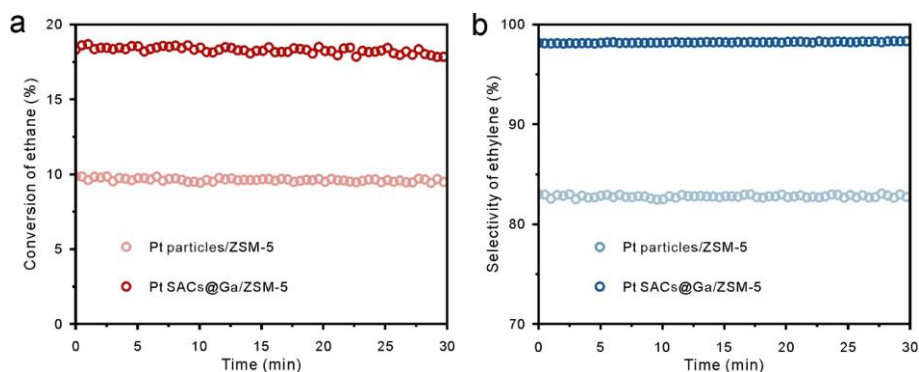


Fig. R3 | Comparison of catalytic performance with and without the introduction of liquid metal. a Conversion of ethane. **b** Selectivity of ethylene.

Changes: To more clearly demonstrate the performance improvements achieved by dispersing Pt in liquid metal Ga, we replaced the performance comparison of pure Ga, pure ZSM-5, and Pt SACs@Ga/ZSM-5 catalysts in Fig. 4b, c with the catalytic performance of Pt particles@ZSM-5 and Pt SACs@Ga/ZSM-5 catalysts. We also provided revised descriptions and analyses for the data in the revised manuscript. The changes are marked in red.

Page 14 in the revised manuscript:

The Pt particles@ZSM-5 catalyst, which is prepared by direct mixing of commercial Pt particles with ZSM-5, exhibits ethane conversion at 9.5% and ethylene selectivity at 82.6% (Figs. 4b, c). Interestingly, the liquid metal dispersed SACs are highly active with ethane conversion at 17.8% and ethylene selectivity at 98.3%. The breakthrough in performance stems from the introduction of liquid metal, which directly alters the state of Pt. By dispersing bulk Pt into single Pt atoms, the utilization of Pt atoms is maximized to achieve high conversion rates. Furthermore, since the size of Pt particles significantly influences catalytic performance, larger Pt nanoparticles typically promote C–C cleavage and deep dehydrogenation side reactions³⁷. In contrast, Pt-based catalysts dispersed at the atomic scale exhibit excellent selectivity for ethylene due to their superior activation of C–H bonds in alkanes, while simultaneously limiting C–C bond cleavage to prevent dehydrogenation reactions. Therefore, when liquid metal is introduced, Pt exists in a single-atom state, enabling the Pt SACs@Ga/ZSM-5 catalyst to achieve nearly 100% selectivity.

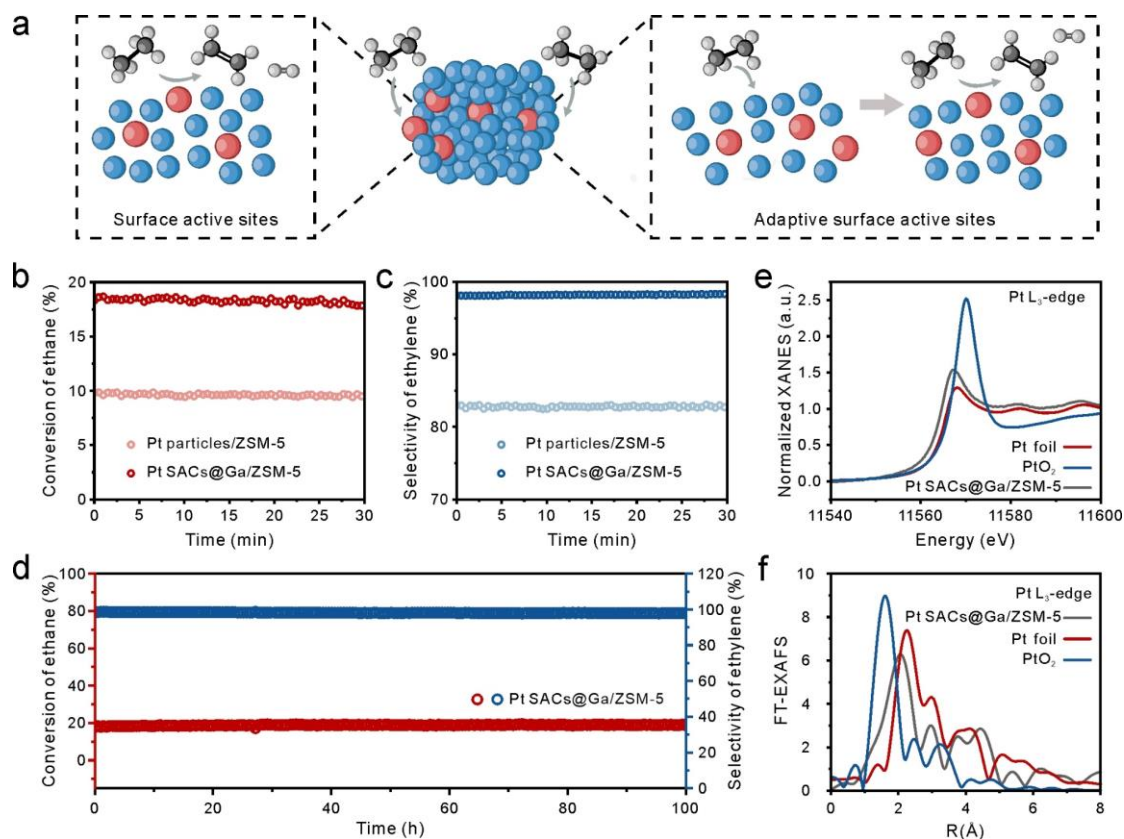


Fig. 4 | Ethane dehydrogenation performance of liquid metal dispersed SACs. (a) Schematic showing the structure evolution of adaptive surface active sites in liquid metal dispersed SACs when exposed to ethane atmosphere. (b, c) Catalytic performances at 650 °C. (d) Long-term stability test. (e) XANES spectra of the liquid metal dispersed SACs after 100 h ethane dehydrogenation taken at Pt L_3 edge. (f) k_3 -weighted FFT spectra in R-space at Pt L_3 edge.

Comment 2: A second major issue is the way the work is positioned with respect to existing literature on SCALMS (Supported Catalytically Active Liquid Metal Solutions) and single-atom/alloy catalysts. Ga–Pt and related molten metal/solid alloy catalysts have been studied for some time, in particular by Peter Wasserscheid and co-workers and others, who have explicitly discussed high-temperature stability and among others Ga-based systems. There are already reports that can reasonably be described as “single atom” or “single site” catalysts embedded in a liquid Ga host on a support, including recent work (e.g. doi:10.1002/cphc.202400651) that addresses exactly the stability and site isolation arguments the authors here put forward. Against this background, the assertion that this manuscript is the first to propose the concept of a supported single-atom catalyst in a molten metal host appears overstated. The results read as an incremental variation within the established SCALMS/SAA/SAC framework rather than as a fundamentally new conceptual advance.

Author reply: Thank you very much for your feedback on our revised manuscript. In terms of the difference in our work and the reported SCALMS, especially the reference (doi:

10.1002/cphc.202400651), we have conducted a comprehensive literature search via SCOPUS. We acknowledged that Prof. Peter and his colleagues, as well as others, have conducted important research in this field. However, the core innovation of our work is different from that of the reported works.

Our work focuses on the severe sintering limitation of SACs at high temperatures in industrial catalysis. By leveraging the fluidity and metal affinity of liquid metal, we realized the ultra-stable SACs at 650 °C for 100 hours. It is worth noting that the high temperature contributes to the dispersion of single atoms in our work rather than the agglomeration sintering phenomenon in reported works, which is owing to the selective breaking and formation of chemical bonds profiting from the inherent metal affinity of liquid metal. Combined with the fluidity of liquid metal, active metal atoms can move but remain confined to the liquid metal as a single-atom state at high temperatures. This finding is of great significance for promoting the industrial application of SACs. **And we provided robust evidence through a combination of theoretical calculations and experimental data** (Fig. 2, Fig. 4 and Supplementary Movs. 1, 2 in the revised manuscript). Additionally, immobilizing Ga–Pt on porous ZSM-5 is mainly to enhance the gas mass transfer efficiency. The focus in this work is to utilize liquid metal to address the bottleneck of agglomeration of SACs.

While **the reference** (doi:10.1002/cphc.202400651) focuses on the Ga–Ni system within SCALMS for selective polymerization of ethylene. The **active center of the catalyst is jointly formed by isolated Ni single atoms and Ga–Ni intermetallic compounds** (e.g., Ga₅Ni, Ga₇Ni₃), which is different from the isolated single atoms as active centers in our work. However, **at elevated temperatures, the low-melting-point Ga₅Ni undergoes a phase transformation** into the high-melting-point Ga₇Ni₃ intermetallic compounds (Fig. R4a). Due to the insolubility Ga₇Ni₃, Ga₇Ni₃ cannot replenish Ni single atoms on the catalyst surface (Fig. R4b), leading to gradual depletion of active sites and subsequent deterioration of catalyst stability (Fig. R4c). Therefore, **the core innovation, the stability of the catalyst, and structure-performance relationship of the reference are totally different from our work.**

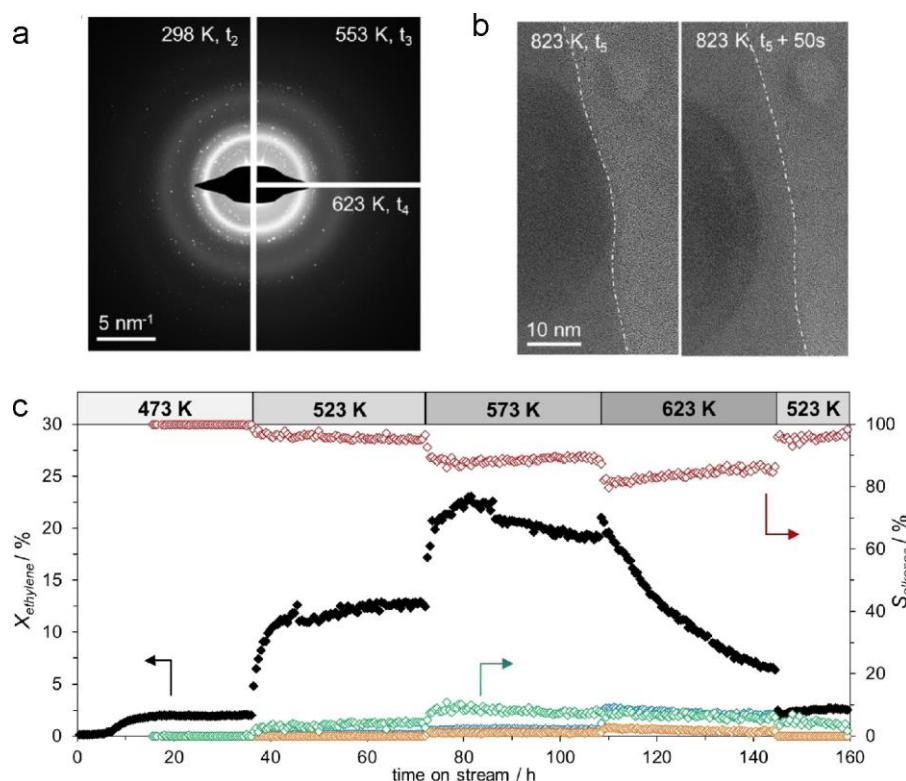


Fig. R4 | Deactivation Mechanism of Ga–Ni SCALMS. **a** Snapshots of in situ SAED at 298 K, 553 K and 623 K. **b** Snapshots of in situ HRTEM taken at 823 K. **c** Ethylene oligomerization using Ga–Ni SCALMS at temperatures ranging from 473 to 623 K.

Furthermore, the **synthetic approaches** of our work and the reported SCALMS by Prof. Peter are different. **And the stability at high temperatures of our catalyst is outstanding among these reported works.**

1) **A fundamental difference in the synthetic approaches.** The approach of Prof. Peter predominantly relies on galvanic replacement driven by differences in reduction potential (*Nat. Chem.* **2017**, 9, 862) or impregnation-reduction (*ChemCatChem*, **2025**, e202402096), whereby the salt of the active metal reacts at the liquid Ga surface. However, galvanic replacement presents certain limitations. One constraint is the uniformity of synthesis. The local electric field, ion concentration, and the distribution of reactants will result in differences in size, composition, distribution, or structure of the active sites formed through galvanic replacement reactions. Another limitation is restricted universality. Electrodeposition requires a potential difference between the two metals, meaning only metals with lower electrode potentials can replace those with higher potentials.

In contrast, our work employs the direct dissolution of metallic Pt into liquid Ga. By leveraging the solvent properties of liquid metal and its strong affinity for Pt, we achieve stable Ga–Pt interactions even at high temperatures, thereby efficiently forming highly uniform catalysts. This highly uniform environment provides an efficient synthetic pathway for producing SACs with a

highly uniform coordination environment. Furthermore, Ga exhibits the favorable affinity and relatively negative mixing enthalpy toward most metallic elements (*Nature* **2023**, 619, 73), suggesting the potential for universal synthesis of liquid metal dispersed SACs.

2) **The superior thermal stability of the catalyst in our work.** By comparing the operating temperatures and time on stream of catalysts in our work versus Prof. Peter's (Tab. R1), it is found that the catalysts in our work reveal a distinct advantage in thermal stability. In the reported work of Prof. Peter that catalysts were synthesized by the galvanic replacement method, active metals are initially enriched at the surface or interface. Besides, high-temperature calcination is typically required in this method to form the final SCALMS structure. At elevated temperatures, **irreversible transformations of the active sites lead to deactivation of the catalyst**. Conversely, our catalyst can maintain a stable single-atom state continuously without undergoing deactivation (evidenced by XAFS in Figs. 4 e, f and theoretical calculation in Fig. 2). This exceptional thermal stability stems from the stable interaction between Pt and Ga at high temperatures, which is a thermodynamically driven process. This has been thoroughly evidenced at the atomic level from thermodynamic and kinetic perspectives in Fig. 2 in our revised manuscript.

Tab. R1 | Summary of Prof. Peter Wasserscheid's work on SCALMS.

No.	Type of catalyst	Method of catalyst synthesis	Catalytic reaction	Temperature (°C)	Stability (h)	Reference
1	In-Pt SCALMS	Wet impregnation	Propane dehydrogenation	500	15	<i>ChemCatChem</i> , 2025 , e202402096
2	Ga ₄₆ Pt/SiO ₂ SCALMS	Galvanic displacement	Propane dehydrogenation	600	15	<i>ChemCatChem</i> , 2025 , e202500176
3	Ga-Ni SCALMS	Galvanic displacement	Ethylene oligomerization	260	120	<i>ChemPhysChem</i> , 2025 , e202400651
4	Ga-Pt SCALMS	Galvanic displacement	Propane dehydrogenation	550	100	<i>ACS Catal.</i> 2025 , 15, 12436
5	Ga-Pt SCALMS	Galvanic displacement	Propane dehydrogenation	550	15	<i>ACS Appl. Nano Mater.</i> 2024 , 7, 24356
6	Ga ₈₄ Pt/Al ₂ O ₃ SCALMS	Galvanic displacement	Dehydrogenation of n-heptane	430	8	<i>React. Chem. Eng.</i> 2024 , 9, 1154
7	Ga-Ni SCALMS	Galvanic displacement	Dry reforming of methane	900	20	<i>Commun. Chem.</i> 2023 , 6, 224
8	Ga-Pt, Ga-Pd, Ga-Rh SCALMS	Galvanic displacement	Dehydrogenation of isobutane	450	15	<i>Chem. Eng. J</i> 2023 , 475, 146081
9	Ga-Pt SCALMS	Galvanic displacement	N-heptane dehydrogenation	430	14	<i>Catal. Sci. Technol.</i> 2023 , 13, 4435
10	Ga-Ni SCALMS (Ga ₆₇ Ni/SiO ₂)	Galvanic displacement	Ethylene oligomerization	260	75	<i>Catal. Sci. Technol.</i> 2021 , 11, 7535
11	Ga-Pt SCALMS	Galvanic displacement	Propane dehydrogenation	550	14	<i>ACS Catal.</i> 2021 , 11, 13423
12	Ga-Pd SCALMS	Galvanic displacement	/	/	/	<i>Catalysts</i> 2021 , 11, 810
13	Ga-Pt SCALMS	Galvanic displacement	Propane dehydrogenation	550	16.7	<i>Faraday Discuss.</i> 2021 , 229, 359
14	Ga ₅₂ Pt/SiO ₂ SCALMS	Galvanic displacement	Dehydrogenation of methylcyclohexane	450	75	<i>ChemCatChem</i> 2020 , 12, 4533
15	Ga-Rh SACLMS	Galvanic displacement	Propane dehydrogenation	500	6	<i>ChemCatChem</i> 2020 , 12, 1085
16	Ga-Rh SACLMS	Galvanic displacement	Propane dehydrogenation	550	15	<i>ACS Catal.</i> 2019 , 9, 9499
17	Ga-Pd SACLMS	Galvanic displacement	Butane dehydrogenation	445	25	<i>Nat. Chem.</i> 2017 , 9, 862

To further substantiate the thermal stability performance of our work, we also compared the catalytic reaction temperatures and time on stream of our work with reported SACs (Tab. R2), where our results remain outstanding.

Tab. R2 | Comparison of thermal stability in this work and other SACs among reported works.

Catalyst	Reaction types	Temperature (°C)	Time (h)	Reference
Pt SACs@Ga/ZSM-5	Ethane dehydrogenation	650	100	This work
Ce-MoTiO _x	Reduction of NO _x with NH ₃	300	240	<i>Appl. Catal.</i> 2025 , 366, 125021
Rh ₁ /Fe-doped CeO ₂	Reduction of NO with CO	400	100	<i>Angew. Chem. Int. Ed.</i> 2025 , 64, e202421218
Co@MFI	Non-oxidative dehydrogenation of ethane	650	40	<i>J. Mater. Chem. A</i> 2022 , 10, 9445
In/Rh@S-1	Propane dehydrogenation	600	5500	<i>Science</i> 2024 , 383, 998
Fe(II)-S-1	Ethane dehydrogenation	550	6	<i>J. Energy Chem.</i> 2024 , 97, 677
Co/S-1	Ethane dehydrogenation	590	102	<i>Chem</i> 2023 , 9, 637
Ni/A-TiO ₂	Propane dehydrogenation	580	5	<i>Chin. J. Catal.</i> 2024 , 57, 105
GaPt/ZSM-5	Oxidative ethane dehydrogenation	650	3	<i>Catal. Commun.</i> 2023 , 176, 106631
Ni/ α -MoC	Methanol dehydrogenation	240	100	<i>Angew. Chem. Int. Ed.</i> 2025 , e202423682
Pt ₄ /Ga	Hydrogenation of acid to alkanes	255	6	<i>Appl. Catal., B</i> 2025 , 365, 124915
Ni ₁ -Mg SAA	Mg hydrogen storage reaction	300	30 cycles	<i>Adv. Mater.</i> 2024 , 36, 2412942
Ni SAs/N-Al ₂ O ₃	Propane dehydrogenation	580	140	<i>Appl. Catal., B</i> 2024 , 347, 123798
Cu-Fe ₂ O ₃	NH ₃ oxidation	300	100	<i>Angew. Chem. Int. Ed.</i> 2024 , 63, e202410457
MOF-808-NaCu	CO ₂ hydrogenation to methanol	275	40	<i>Natl. Sci. Rev.</i> 2024 , 11, nwae114
Co-Ba/Y ₂ O ₃	Ammonia decomposition	500 – 600	110 h at 550 °C and 240 h at 500 °C	<i>Angew. Chem. Int. Ed.</i> 2025 , 64, e202416195
Pt/CeO _x /SiO ₂	H ₂ reduction	300	10	<i>Nature</i> 2022 , 611, 284
Cu ₁ / γ -Al ₂ O ₃	Methane to methanol	200	12	<i>Appl. Catal., B</i> 2025 , 361, 124648
Ru-(H)L(i-C ₃) ₂ /AC	Acetylene hydrochlorination	170	500	<i>ACS Catal.</i> 2025 , 15, 3085
RuNi SAA	Hydrogenation of nitroarenes	60	3	<i>Nat. Commun.</i> 2022 , 13, 3188
Pd ₁ Ni/SiO ₂	Hydrogenation of acetylene	80	12	<i>Appl. Catal. A-Gen.</i> 2023 , 662, 119288
Ru ₁ CoNP	Synthesis of piperidines and pyridine from furfural	220	4	<i>Nat. Commun.</i> 2023 , 14, 6329
RuNi SAA	CO ₂ hydrogenation	400	140	<i>Angew. Chem. Int. Ed.</i> 2025 , 64, e202412835
Ni ₁ /CoCeO ₂	Dry reforming of methane	800	150	<i>Appl. Catal., B</i> 2022 , 314, 121516.
Co ₁ /V ₂ O ₅	Acceptorless dehydrogenative coupling of benzyl alcohol	359	0.5	<i>Small</i> 2023 , 19, 2207941
Pt-NiCe@SiO ₂	Dry reforming of methane	550	120	<i>ACS Catal.</i> 2021 , 11, 8247
Mn/V/Al ₂ O ₃ and Mn/Ni/ZrO ₂	Oxidative dehydrogenation of propane	300 – 500	25	<i>ACS Catal.</i> 2024 , 14, 16698
Fe-ZIF-8-800	Hydrogenation of furfural	60 – 120	6	<i>Nat. Commun.</i> 2023 , 14, 6666
Rh ₁ /SnO ₂	Ethylene hydroformylation	150	85	<i>Angew. Chem. Int. Ed.</i> 2023 , 62, e202214048
Mo/NiFe ₇₁ -V	Hydrodeoxygenation of lignin to naphthene	240	3	<i>Chem. Eng. J</i> 2025 , 505, 159303
Ni-SAs/N-CNTs	Reverse water-gas shift reaction	500	100	<i>ACS Catal.</i> 2023 , 13, 7132
Co@Mo-S-1	Propane dehydrogenation	600	72	<i>J. Am. Chem. Soc.</i> 2024 , 146, 8939
PtSn/nano-Al ₂ O ₃	Propane dehydrogenation	600	40	<i>Chin. J. Catal.</i> 2023 , 48, 164
B-Ni in MnO ₂	Methane catalytic combustion	500	32	<i>Nat. Commun.</i> 2025 , 16, 1008

In summary, through comparative analysis, we found that our work exhibits a fundamental difference in core innovation, synthetic approach, and the stability of the active centers compared to previously reported work. It is not merely an incremental advancement of prior research but represents an entirely new concept. We present a new approach to achieve highly thermal-stable SACs by dispersing metal Pt into single atoms by liquid metal Ga, which is driven by the stable Pt–Ga interaction even at elevated temperatures. This is a thermodynamically driven process, and the corresponding mechanism was thoroughly evidenced from both thermodynamic and kinetic perspectives.

Changes: For highlighting the advantages of our work, the stability of the reported SACs and ours has been summarized as Supplementary Tab. 1 in the revised manuscript.

Comment 3: *In addition, the experimental scope is rather narrow for a paper targeting Nature Communications. The study focuses on a single reaction (ethane to ethylene) under relatively limited conditions, with a small set of catalyst formulations and without an in-depth kinetic analysis (e.g. an estimate of turnover frequencies per Pt vs. Ga, reaction orders) or operando characterization that would firmly establish the structure and environment of the proposed Pt single atoms under working conditions. Long-term stability, deactivation pathways, and structure-performance relationships are not explored to the depth one would expect for the broad readership of such a journal. The resulting dataset is essentially a proof of concept confined to one reaction and one composition space, which may be more appropriate for a specialized catalysis journal.*

Author reply: Thank you very much for your constructive comments, which have significantly helped us to enhance the depth and breadth of our work. In response to the concerns regarding the scope of our experiments, we have supplemented the systematic supplementary studies and discussions to analyze the applicability of the reaction system of our materials, reaction kinetic analysis, the generalizability of our findings, and the long-term stability, deactivation pathways, and structure-performance relationships. The detailed clarification is as follows:

(1) **Regarding the catalysis reaction applicability of this catalyst**, EDH was chosen as a proof of concept, whose catalytic stability is highly sensitive to the dispersion state and coordination environment of the active metals (*Angew. Chem. Int. Ed.* **2024**, *63*, e202409784). Usually, when the size of the active metals is increased, the coordination environment of each active atom is highly non-uniform. This will typically promote C–C cleavage and deep dehydrogenation side reactions, which severely hinder the stability of the catalyst (*ACS Catal.* **2020**, *10*, 12932). Therefore, **EDH serves as an exceptionally sensitive probe reaction for revealing the high-temperature stability of our liquid-metal-supported SACs.**

The core innovation of our work lies **in addressing the problem of severe sintering of SACs operated at high-temperature conditions in thermocatalysis by leveraging the fluidity and metal affinity of liquid metal**. Profiting from the superior metal affinity of Ga with Pt, Pt–Pt bonds are broken at elevated temperatures, and subsequently the mobile Pt atoms are captured by Ga to form the stable Pt–Ga interaction, accomplishing the highly efficient dispersion of Pt single atoms. The fluidity of liquid metal and the stable Pt–Ga interaction ensure that the Pt atoms can remain dynamically scattered throughout the whole liquid metal dispersion system. Additionally, liquid metal dispersed single atoms can self-update onto the catalysis interface continuously, promoting catalytic reactions at the interface. Hence, these liquid metal-dispersed SACs manifested remarkable thermal durability and excellent ethylene selectivity for EDH.

Profiting from the favorable stability of the synthesized SACs, this approach is anticipated to be extendable to other thermocatalysis reactions, such as CO₂ hydrogenation, reverse water-gas shift reactions, selective hydrogenation, coupling reactions, alkene epoxidation, and Fenton-like reactions, etc. These reactions occur under heating or even high-temperature conditions, imposing higher demands on the thermal stability of catalysts. And multiple side reactions are prone to occur, necessitating a highly uniform coordination environment around the catalytic active site to minimize or suppress the occurrence of these side reactions. Our strategy holds promise for achieving high stability and selectivity in these reactions. Accordingly, applying these catalysts to broader reaction systems will be another major focus of our future work.

(2) In terms of the catalyst formulations, the concept of liquid metal-dispersed SACs can be applied to **synthesizing Rh-based SACs (Rh SACs@Ga/ZSM), and this catalyst also demonstrates the improved EDH performance**. In response to reviewer requests, we selected the GaRh system to demonstrate the universality of this approach for synthesizing liquid metal-dispersed SACs. XAFS analysis confirmed the formation of distinct Ga–Rh bonds (Fig. R5). Furthermore, the synthesized Rh SACs@Ga/ZSM was employed in the EDH. The Rh SACs@Ga/ZSM-5 catalyst achieved an ethane conversion rate of 16.9% and an ethylene selectivity of 91.3% (The atomic ratio of Ga and Rh in this catalyst is 99.8 : 0.2). While the Rh particles/ZSM-5 exhibited only 13.2% conversion and a mere 81.4% ethylene selectivity under the same amount of Rh (Fig. R6). Thus, the synthesis method in our work exhibits certain applicability and can be extended to other catalyst formulations.

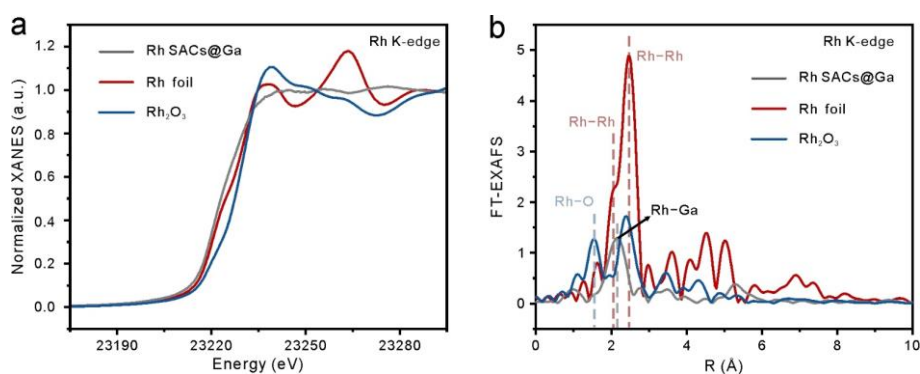


Fig. R5 | Normalized XANES spectra at the Rh K-edge (a) and FT Rh K-edge EXAFS spectra in R space (b) of liquid metal dispersed Rh SACs.

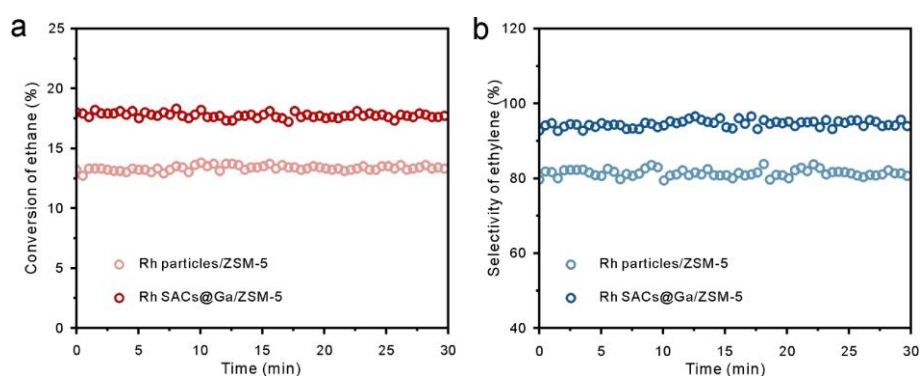


Fig. R6 | Catalyst performance of liquid metal dispersed Rh SACs and Rh particles. (a) Conversion of ethane. (b) Selectivity of ethylene.

(3) As for the catalytic test conditions and in-depth **kinetic analysis**, we have incorporated a **comprehensive intrinsic kinetic analysis** within the currently prevalent temperature range for **alkane dehydrogenation** (*Science* **2024**, 383, 998; *Chem* **2023**, 9, 637; *J. Am. Chem. Soc.* **2024**, 146, 8939), including the turnover frequency (TOF), apparent activation energy (E_a), and reaction order. This not only further clarifies the advantages of liquid metal dispersed SACs in reaction kinetics, but also verifies their catalytic activity at different temperatures and different reactant concentrations. Through calculation, the TOF of the SACs is approximately 3.69 h⁻¹. **The activation energy of Pt SACs@Ga/ZSM-5 for the EDH is about 102.44 kJ mol⁻¹, which is 34.74 kJ mol⁻¹ lower than that of the Pt particles/ZSM-5 catalyst.** Furthermore, the ethane reaction order is comparable to that of the Pt particles/ZSM-5 catalyst, with no significant difference in ethane adsorption.

a) TOF Calculation. The turnover frequency (TOF) for ethylene production was calculated using the following equation:

$$TOF_{C_2H_4} = \frac{F_{C_2H_6} \times Conv_{C_2H_6} \times Sel_{C_2H_4}}{m_{cat} \times \omega_{Pt} \times D_{Pt}}$$

where $F_{C_2H_6}$, $Conv_{C_2H_6}$ and $Sel_{C_2H_4}$ represent the flow rate and conversion of ethane, the selectivity of ethylene, respectively. And m_{cat} represents the total mass of the catalyst. ω_{Pt} represents the total mass fraction of active metal Pt in the catalyst, which is determined by ICP-OES. D_{Pt} represent the dispersion of the active Pt on the surface of the catalyst. Based on the measurement of CO pulse adsorption, D_{Pt} is approximately 13.96%. Based on these data, the calculated TOF for ethylene over our catalyst is 3.69 h^{-1} .

b) Determination of Activation Energy and Reaction Order. To investigate the kinetic behavior of the catalyst for EDH, we measured reaction rates under conditions designed to minimize transport limitations. Specifically, we limited the conversion to 5 ~ 15% while ensuring ethylene selectivity remained above 90% by systematically varying the catalyst amount. By analyzing the temperature and pressure dependence of these rates, we extracted the activation energy and reaction order for the EDH.

We measured the ethane conversion in a temperature range between 883 K to 933 K and fitted the data according to the Arrhenius equation to calculate the activation energy for EDH. The resulting activation energy for the Pt SACs@Ga/ZSM-5 catalyst is $102.44 \text{ kJ mol}^{-1}$ (Fig. R7a). For comparison, we also calculated the apparent activation energy of the Pt particles/ZSM-5 in the EDH reaction. However, due to the poor selectivity of this catalyst, side reactions will inevitably occur. Therefore, we only controlled its conversion rate to be within the range of 5 ~ 15%. The activation energy for the extreme reaction is $137.18 \text{ kJ mol}^{-1}$ (Fig. R7b). The significantly reduced activation energy indicates a greatly decreased energy barrier that needs to be overcome for the reaction to proceed. And the single-atom Pt formed in the liquid Ga can more effectively weaken the C–H bond of ethane than Pt particles. Therefore, the Pt SACs@Ga/ZSM-5 catalyst exhibits more excellent catalytic performance.

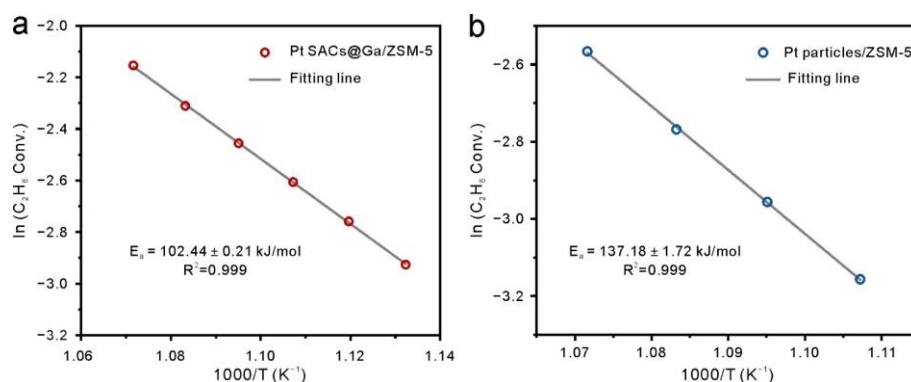


Fig. R7 | Arrhenius plots for EDH reaction. (a) Pt SACs@Ga/ZSM-5. (b) Pt particles/ZSM-5.

The ethane reaction order was determined under the condition where the total flow rate of the

mixture of ethane and argon was 20 sccm, and the partial pressure of ethane was tested to be within the range of 7.5 ~ 17.5%, and the temperature was constant at 923 K. The ethane reaction order for Pt SACs@Ga/ZSM-5 catalyst was determined to be 1.097 (Fig. R8a), while the reaction order for Pt particles/ZSM-5 catalyst is approximately 0.94 (Fig. R8b). The similar reaction order indicates that both are comparable in their adsorption of ethane.

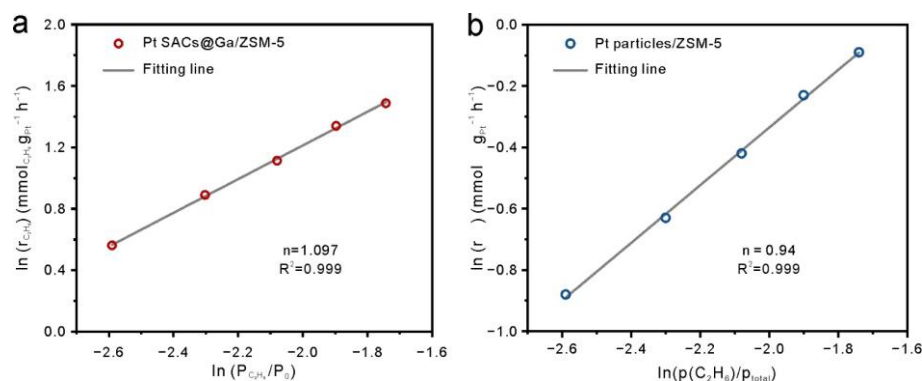


Fig. R8 | Effect of C₂H₆ partial pressure on product distribution in EDH reaction. a Pt SACs@Ga/ZSM-5. **b** Pt particles/ZSM-5.

(4) In terms of the **operando experiments**, we attempted operando XAFS experiments. However, due to the content of Pt being only 0.3 wt% in the Pt SACs@Ga/ZSM used for EDH reactions compared to that of liquid metal Ga, the fluorescence signal intensity of Pt *L*₃ edge is very weak (The atomic ratio of Ga and Pt in this catalyst is 99.8 : 0.2). Furthermore, the increased fluidity of liquid metal at elevated temperatures will lead to severe scattering, which seriously interferes with the weak oscillation signals related to the chemical state and local structure of elements (i.e., EXAFS oscillation), reducing the signal-to-noise ratio and data quality. Therefore, the collection of the XAFS signal in high-temperature operating conditions is extremely challenging.

To investigate the structure and environment of the proposed Pt single atoms under working conditions, we employed ***in situ* XRD and XAFS measurements**. The results indicate that Pt particles spontaneously disperse in the liquid metal to form a stable Ga–Pt interaction at high temperatures. During the *in situ* XRD measurements, initially, the characteristic peaks of Pt metals can be apparently observed, which are located at 39.8° and 46.3°. With the temperature increasing, the characteristic peaks of Pt metals were gradually weakened and disappeared subsequently, proving the dispersion of Pt particles (Fig. R9). Besides, combined with *ex situ* synchrotron XAFS measurements before and after EDH tests for 100 hours, the results indicate that Pt maintains atomic-level dispersion after prolonged thermal stability testing (Fig. R10).

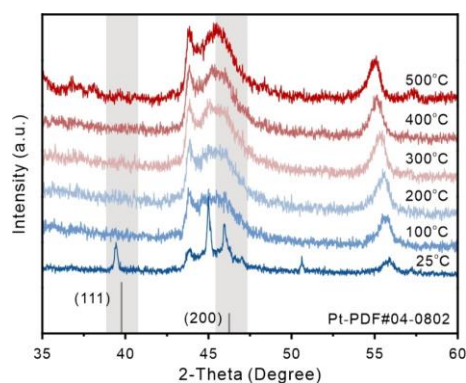


Fig. R9 | In situ XRD patterns of Pt@Ga on ZSM-5 in the process of heating up.

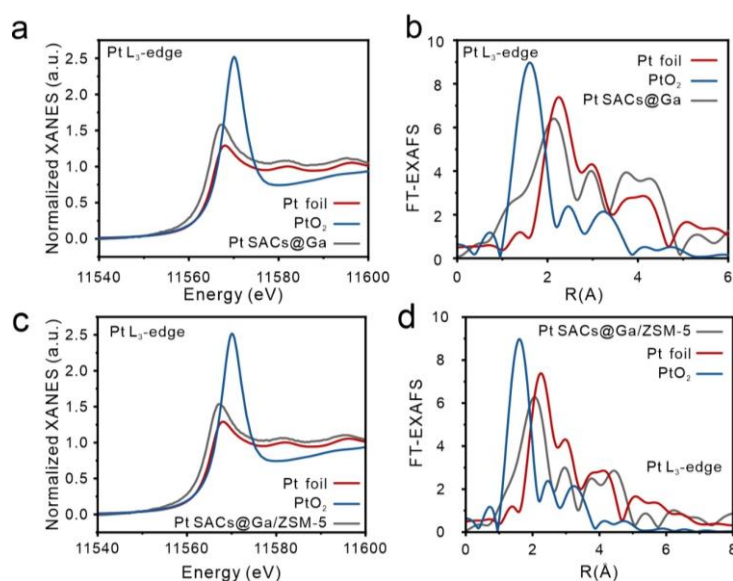


Fig. R10 | Normalized XANES spectra at the Pt L_3 -edge and FT Pt L_3 -edge EXAFS spectra in R space of liquid catalyst before and after the ethane dehydrogenation.

(5) To address the concerns about **long-term stability, deactivation pathways, and structure – performance relationships**, we conducted a 100-hour on-stream stability test under reaction conditions, and the results show that both ethane conversion and propylene selectivity remain nearly unchanged without obvious deactivation (Fig. R11). Moreover, XAFS characterization of Pt SACs@Ga/ZSM-5 catalyst after the EDH for 100 hours confirms that Pt still maintains an atomic dispersion state in liquid metal, with no formation of Pt clusters or nanoparticles, and their local coordination environment is consistent with that of the fresh catalyst (Fig. R10). Based on most previous reports, it is widely believed that single atoms readily sinter into particles at high temperatures, leading to decreased activity, while low selectivity can cause deep cracking side reactions (*Acc. Chem. Res.* **2013**, *46*, 1720; *Phys.Chem.Chem.Phys.*, **2017**, *19*, 30513; *Chem. Soc. Rev.*, **2018**, *47*, 4314; *Nat. Catal.* **2025**, *8*, 1038). The stable catalytic performance of Pt

SACs@Ga/ZSM-5 in this work indicates that the metal-metal interaction effectively stabilizes the Pt single atoms, completely suppressing the main deactivation pathways of Pt-based catalysts, such as sintering and coke deposition reported in previous studies.

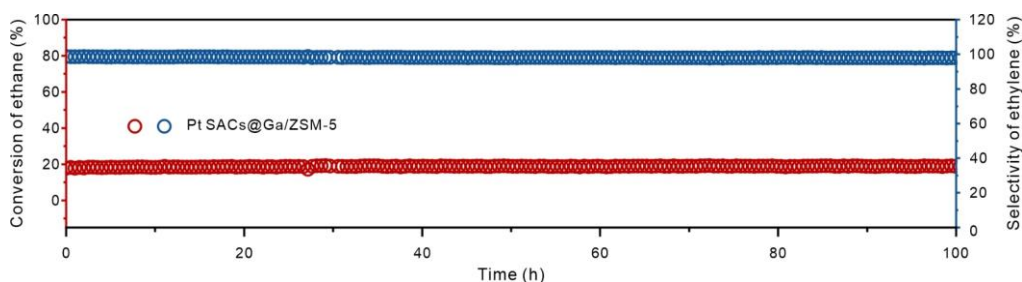


Fig. R11 | Long-term stability test for ethane dehydrogenation performance of liquid metal dispersed SACs.

The SACs maintain high performance and excellent thermal stability at elevated temperatures without deactivation. The fundamental reason lies in the thermodynamically favorable process of liquid Pt dispersing within liquid metal at high temperatures. We elucidate the mechanism of this stability through theoretical calculations

a) Thermodynamic analysis reveals that Pt is more likely to exhibit as the single-atom dispersion state within Ga from the perspective of thermodynamics. The dispersed Pt atoms generally exhibit lower total energy compared to the aggregated Pt clusters within the simulated 10 ps (Fig. R12).

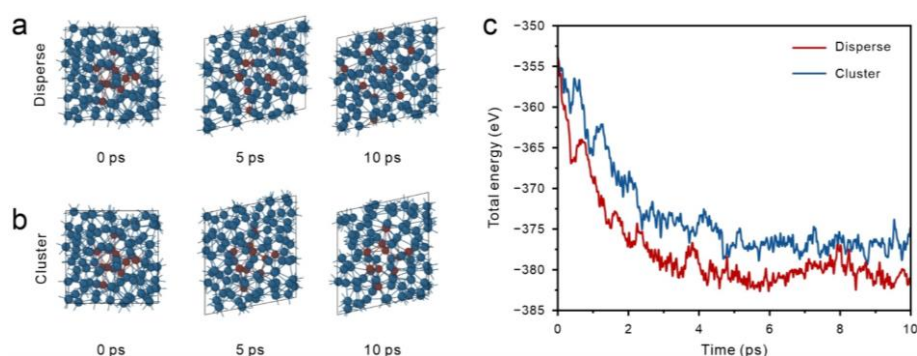


Fig. R12 | Theoretical calculations of heterogeneous active metal atoms monodispersed in liquid Ga. a, b The extracted 0 ps, 5 ps, and 10 ps snapshots from AIMD trajectories of both clustered (a) and dispersed (b) Pt₁₀ in liquid Ga. **c** Total energy versus time curves for the Ga₉₀Pt₁₀ system containing clustered Pt₁₀ and dispersed Pt₁₀.

b) Dynamical simulations of Pt atom diffusion within Ga demonstrate that Pt atoms are resistant to aggregation. The actual existence state of Pt atoms in liquid Ga was studied using AIMD

simulations on the $\text{Ga}_{98}\text{Pt}_2$ system (Pt dimer embedded in Ga_{98}) at 923 K. The 10 ps NVT ensemble simulations (Fig. R13) demonstrate that the Pt–Pt interatomic distance progressively increases from 4.18 Å to 11.63 Å, significantly exceeding the typical Pt–Pt bond length range of 2.65–2.79 Å (*J. Am. Chem. Soc.* **1978**, *100*, 2074). These results confirm that Pt atoms maintain stable single-atom states under high temperature.

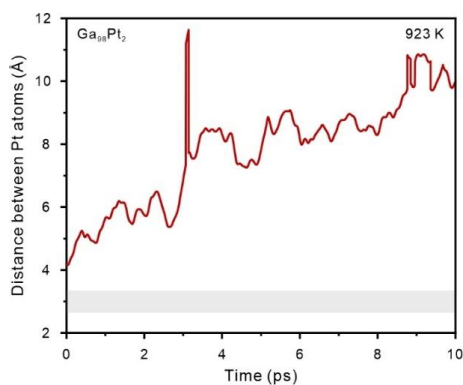


Fig. R13 | Time evolution of Pt–Pt interatomic distance in the $\text{Ga}_{98}\text{Pt}_2$ system under NVT ensemble simulation.

Changes: We have incorporated testing of a comprehensive intrinsic kinetic analysis and verification of method universality into the revised manuscript, with supplementary information provided in both the main text and Supplementary Information (Supplementary Figs. 27 ~ 29).

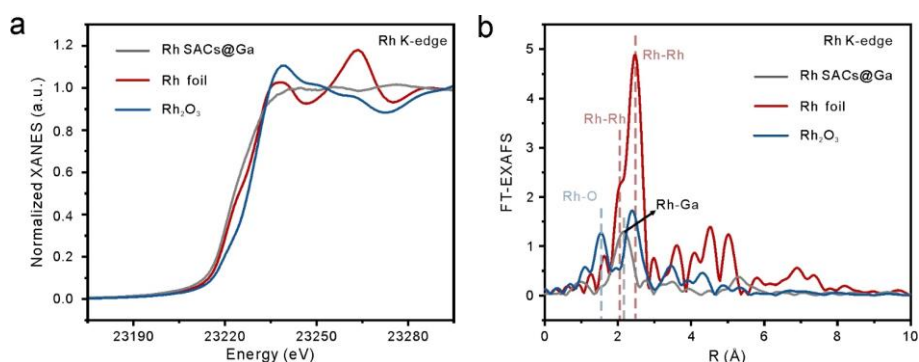
Page 14–15 in revised manuscript:

To verify the universality of the strategy that liquid metals can disperse metals into single atoms, we extended the synthesis method to other systems and successfully synthesized liquid metal

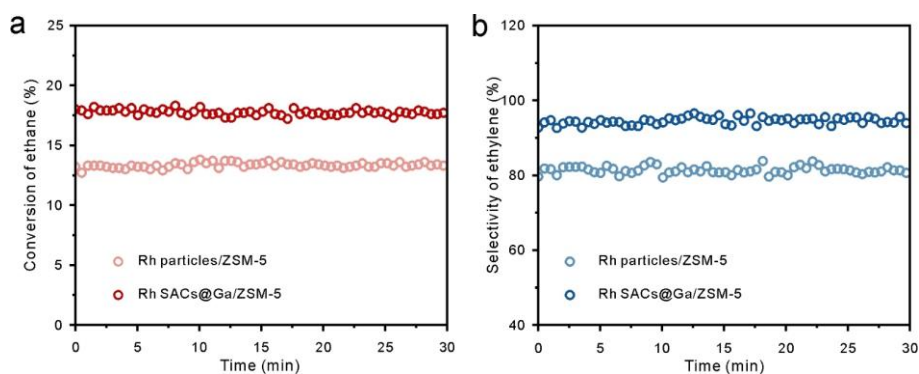
dispersed Rh SACs (Supplementary Fig. 27). Moreover, Rh SACs@Ga/ZSM-5 exhibits significantly improved catalytic performance compared to Rh particles/ZSM-5 in EDH (Supplementary Fig. 28).

To gain a deeper understanding of the effects of liquid metal dispersed SACs and nano catalysts on the dehydrogenation reaction of ethane, we conducted kinetic studies on both types of catalysts. We measured the ethane conversion rates in a temperature range of 883 ~ 933 K and fitted the data according to the Arrhenius equation to calculate the activation energy for EDH. The apparent activation energies for Pt SACs@Ga/ZSM-5 and Pt particles/ZSM-5 are 102.44 kJ mol⁻¹ and 137.18 kJ mol⁻¹, respectively (Supplementary Fig. 29). The significantly reduced activation energy indicates a greatly decreased energy barrier that needs to be overcome for the reaction to proceed. And the single Pt atom dispersed in the liquid Ga can more effectively weaken the C–H bond of ethane than Pt particles. Therefore, the Pt SACs@Ga/ZSM-5 catalyst exhibits more excellent catalytic performance.

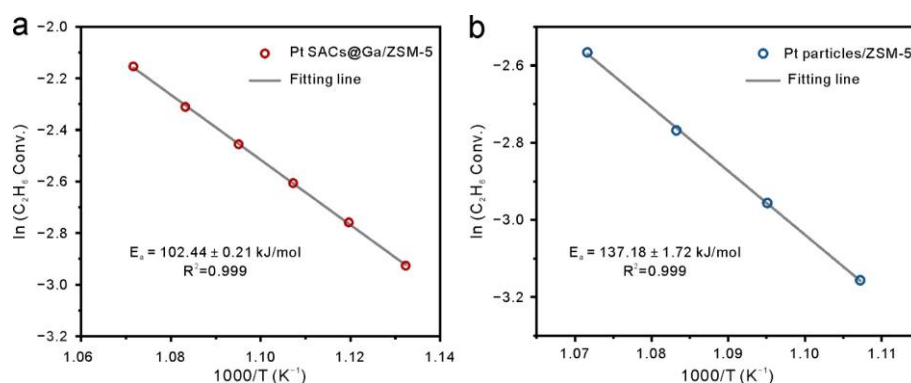
Page 34–36 in revised supplementary information:



Supplementary Fig. 27 | Characterization of liquid metal dispersed Rh SACs. a Normalized XANES spectra at the Rh K-edge. **b** FT Rh K-edge EXAFS spectra in R space of liquid catalyst with phase correction.



Supplementary Fig. 28 | Ethane dehydrogenation performance of liquid metal dispersed Rh SACs. a Conversion of ethane. **b** Selectivity of ethylene.



Supplementary Fig. 29 | Arrhenius plots for EDH reaction. a Pt SACs@Ga/ZSM-5. **b** Pt particles/ZSM-5.

To investigate the kinetic behavior of the catalyst for ethane dehydrogenation, it is necessary to minimize the limitations of material transport as much as possible. Specifically, we limited the conversion to 5 ~ 15% while ensuring ethylene selectivity remained above 90% by systematically varying the catalyst amount. However, due to the poor selectivity of Pt particles/ZSM-5, side reactions are inevitable. Therefore, we only controlled its conversion rate to be within the range of 5 ~ 15%.

Comment 4: *In summary, the manuscript is methodologically solid and the topic is of interest to the catalysis community, but the incremental nature of the contribution, the limited experimental scope, the overstatement of conceptual novelty compared to the established SCALMS and SAC/SAA literature, and the unconvincing treatment of the role of Pt versus Ga prevent me from viewing this as a strong candidate for publication in Nat. Commun. I therefore do not recommend publication of this manuscript in its current form.*

Author reply: Thank you very much for your feedback on our revised manuscript. Our work proposed a solution for agglomeration of SACs in industrial thermal catalytic processes. This finding is substantiated through theoretical calculations and XAFS analysis before and after catalytic reaction. It fundamentally differs from the innovations reported in previous SCALMS, SAC, and SAA studies.

Regarding the roles of Pt and Ga, we re-conducted tests following your suggestion and found Pt to be the primary active site, while Ga does not directly contribute to the catalytic conversion. However, Ga enables Pt particles to disperse into Pt single atoms, achieving breakthroughs in selectivity and conversion. We have revised the data and descriptions in the paper accordingly. Kinetic analysis further revealed that using liquid metal to disperse Pt from particles into single atoms significantly reduces the activation energy for the EDH reaction.

To address limitations in the experimental scope, we extended the system from Pt to Rh and confirmed the universality of the synthesis strategy via XAFS. Catalytic activity was also validated through the EDH reaction, demonstrating superior selectivity and conversion rates compared to Rh particles. We believe this approach has the potential to be extended to broader systems.

We are delighted to hear that you found the new version to be a significant improvement and that your previous concerns have been fully addressed. We appreciate your recommendation for publication.

To Referee #5

We appreciate the referee's highly positive evaluation of our manuscript. We have made a point-by-point response to your comments and carefully revised the manuscript as you suggested. For your reference, please find our revisions marked in red.

Overall remark: *This manuscript reports a liquid metal catalytic system in which Pt dissolved in liquid Ga exhibits **exceptionally stable performance** in ethane dehydrogenation. Although the concept of using liquid metals in catalysis is not entirely new, **the remarkable long-term stability demonstrated here is still appealing. The theoretical analysis, especially the MD simulations, on the thermodynamic stability of the single-atom Pt species is also insightful. In my view, this work is suitable for publication in Nature Communications.***

Author reply: We are grateful for your great summary and approval of the novelty and significance of our work. According to your suggestions, we have carefully revised our manuscript. For your reference, the revisions made in the revised manuscript are highlighted in red color and the detailed point-by-point replies are listed below.

Comment 1: *In Fig. 2e, the Pt–Pt interaction appears stronger than that of Pt–Ga. Why, then, does platinum remain atomically dispersed in liquid gallium? The authors argue that “The calculated bond dissociation energies show a clear ranking: $E(\text{Pt–Pt})$ (107.9 kcal/mol) > $E(\text{Ga–Pt})$ (74.9 kcal/mol) > $E(\text{Ga–Ga})$ (32.5 kcal/mol). Crucially, the formation of two Ga–Pt bonds (149.8 kcal/mol) is energetically favored over either maintaining a Pt–Pt bond (107.9 kcal/mol) or the combined Pt–Pt and Ga–Ga bonding (140.4 kcal/mol).” Yet, this explanation is not entirely convincing, because the numbers of the formed and dissociated bonds seem over simplified for the liquid phase.*

Author reply: Thank you for your constructive comment. As you have pointed out, compared to the actual liquid Ga environment, merely considering the reaction $\text{Pt}_2 + \text{Ga}_2 \rightarrow 2\text{PtGa}$ is indeed a simplified approach. Actually, to better account for the liquid Ga environment, we have calculated the formation energies and reaction energy barriers for embedding Ga atoms into Pt_2 , $(\text{Ga}_6\text{Pt})_2$, and $(\text{Ga}_7\text{Pt})_2$ clusters, which leads to the formation of Pt_2Ga , $(\text{Ga}_6\text{Pt})_2\text{Ga}$, and $(\text{Ga}_7\text{Pt})_2\text{Ga}$ clusters. The results (Fig. R1) indicate that this process is both thermodynamically and kinetically feasible at 923 K.

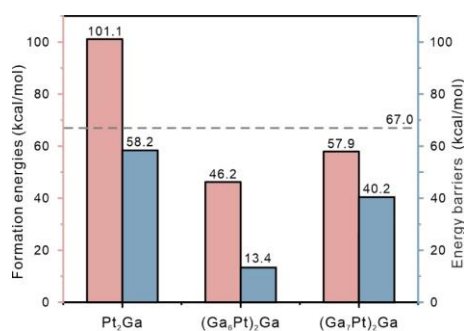


Fig. R1 | The formation energies and energy barriers associated with Ga atom insertion into Pt–Pt bonds in Pt₂Ga, (Ga₆Pt)₂Ga, and (Ga₇Pt)₂Ga molecular systems. The rough estimation suggests that the reaction can overcome an energy barrier of 67 Kcal mol^{−1} at 923 K.

To clarify the origin of this thermodynamic stability, the energy difference between breaking the Ga–Ga and Pt–Pt bonds and forming two Ga–Pt bonds during this process was comparatively analyzed. As your comment highlighted the oversimplification of the calculated model, we further **incorporated the molecular structures of Pt₂, (Ga₁Pt)₂, (Ga₂Pt)₂, (Ga₃Pt)₂, (Ga₄Pt)₂, and (Ga₅Pt)₂ into our analysis** (Fig. R2). Based on the optimized geometries, the Pt–Pt bond lengths in the stable structures of (Ga₄Pt)₂ and (Ga₅Pt)₂ are 3.101 Å and 3.380 Å, respectively. These values significantly exceed 1.2 times the Pt–Pt bond length of 2.377 Å in Pt₂, indicating that the Pt–Pt bonds in the stable configurations of (Ga₄Pt)₂ and (Ga₅Pt)₂ can be considered effectively broken. Therefore, we calculated the bond energy difference ΔE_b ($\Delta E_b = E_{(Ga_nPt)_2(Pt-Pt)} + \bar{E}_{Ga_n(Ga-Ga)} - 2E_{Ga_{n+1}Pt(Ga-Pt)}$) for the Pt₂, (Ga₁Pt)₂, (Ga₂Pt)₂, and (Ga₃Pt)₂ molecular clusters. Here, $\bar{E}_{Ga_n(Ga-Ga)}$ represents the average Ga–Ga bond energy in the Ga_n molecular clusters, $E_{(Ga_nPt)_2(Pt-Pt)}$ denotes the Pt–Pt bond energy in the (Ga_nPt)₂ molecular clusters, and $E_{Ga_{n+1}Pt(Ga-Pt)}$ refers to the Ga–Pt bond energy in the Ga_{n+1}Pt molecular clusters. All bond energy calculations were corrected for basis set superposition error (BSSE). The results, as shown in Fig. R3, indicate that **the ΔE_b values for the Pt₂, (Ga₁Pt)₂, (Ga₂Pt)₂, and (Ga₃Pt)₂ molecular clusters**, evaluated using different computational methods (B3LYP, MN15L, and M06L), **are all greater than zero**. This demonstrates that in these molecular systems, **the Ga atoms involved in bond-breaking readily disrupt the Pt–Pt bonds, leading to the formation of stable Pt–Ga–Pt bridging units, thereby promoting the single-atom dispersion of Pt in liquid Ga.**

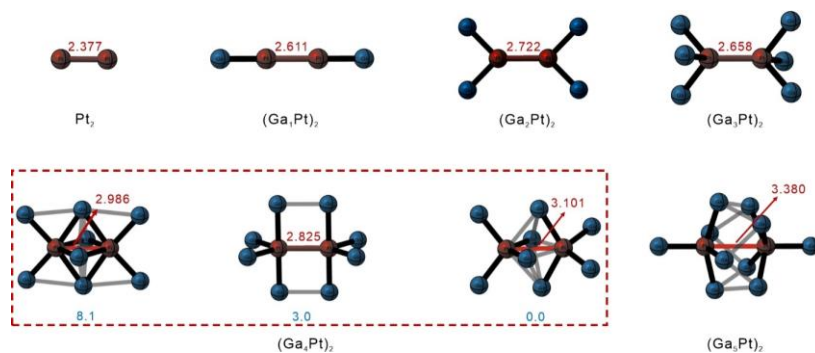


Fig. R2 | The optimized structures of the molecular systems Pt_2 , $(\text{Ga}_1\text{Pt})_2$, $(\text{Ga}_2\text{Pt})_2$, $(\text{Ga}_3\text{Pt})_2$, $(\text{Ga}_4\text{Pt})_2$, and $(\text{Ga}_5\text{Pt})_2$. The red numbers denote the Pt–Pt bond lengths in angstroms (Å), and the blue numbers denote the relative energies (in kcal mol^{-1}) of the three isomers of $(\text{Ga}_4\text{Pt})_2$.

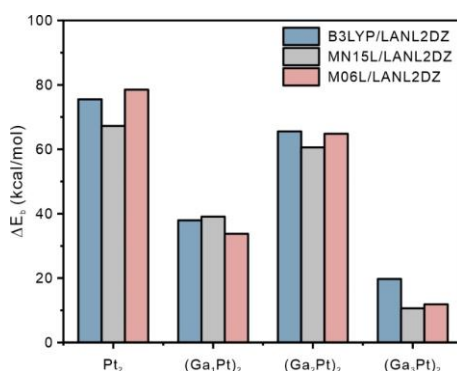


Fig. R3 | The bond energy difference (ΔE_b). $\Delta E_b = E_{(\text{Ga}_n\text{Pt})_2(\text{Pt-Pt})} + \bar{E}_{\text{Ga}_n(\text{Ga-Ga})} - 2E_{\text{Ga}_{n+1}\text{Pt}(\text{Ga-Pt})}$ was calculated using different computational methods (B3LYP, MN15L, and M06L) for the Pt_2 , $(\text{Ga}_1\text{Pt})_2$, $(\text{Ga}_2\text{Pt})_2$, and $(\text{Ga}_3\text{Pt})_2$ molecular clusters.

Changes: In the revised version, we replaced the original simplified model ($\text{Pt}_2 + \text{Ga}_2 \rightarrow 2\text{PtGa}$), incorporated the molecular structures of Pt_2 , $(\text{Ga}_1\text{Pt})_2$, $(\text{Ga}_2\text{Pt})_2$, $(\text{Ga}_3\text{Pt})_2$, $(\text{Ga}_4\text{Pt})_2$, and $(\text{Ga}_5\text{Pt})_2$ into our analysis. And we calculated the bond energy difference to assess the stability of the Ga–Pt bond. The changes are marked in red in the revised manuscript.

Page 8 in the revised manuscript:

To fundamentally understand the thermodynamic preference for Pt mono-dispersion in liquid Ga, we systematically calculated the bond energy difference ΔE_b ($\Delta E_b = E_{(\text{Ga}_n\text{Pt})_2(\text{Pt-Pt})} + \bar{E}_{\text{Ga}_n(\text{Ga-Ga})} - 2E_{\text{Ga}_{n+1}\text{Pt}(\text{Ga-Pt})}$) only for the Pt_2 , $(\text{Ga}_1\text{Pt})_2$, $(\text{Ga}_2\text{Pt})_2$, and $(\text{Ga}_3\text{Pt})_2$ molecular clusters (Fig. 2e and Supplementary Figs. 6 ~ 11). Here, $\bar{E}_{\text{Ga}_n(\text{Ga-Ga})}$ represents the average Ga–Ga bond energy in the Ga_n molecular clusters, $E_{(\text{Ga}_n\text{Pt})_2(\text{Pt-Pt})}$ denotes the Pt–Pt bond energy in the $(\text{Ga}_n\text{Pt})_2$ molecular clusters, and $E_{\text{Ga}_{n+1}\text{Pt}(\text{Ga-Pt})}$ refers to the Ga–Pt bond energy in the Ga_{n+1}Pt molecular

clusters. The results indicate that the ΔE_b values for the Pt_2 , $(Ga_1Pt)_2$, $(Ga_2Pt)_2$, and $(Ga_3Pt)_2$ molecular clusters, evaluated using different computational methods (B3LYP, MN15L, and M06L), are all greater than zero. This demonstrates that in these molecular systems, the Ga atoms involved in bond-breaking readily disrupt the Pt–Pt bonds, leading to the formation of stable Pt–Ga–Pt bridging units, thereby promoting the single-atom dispersion of Pt in liquid Ga.

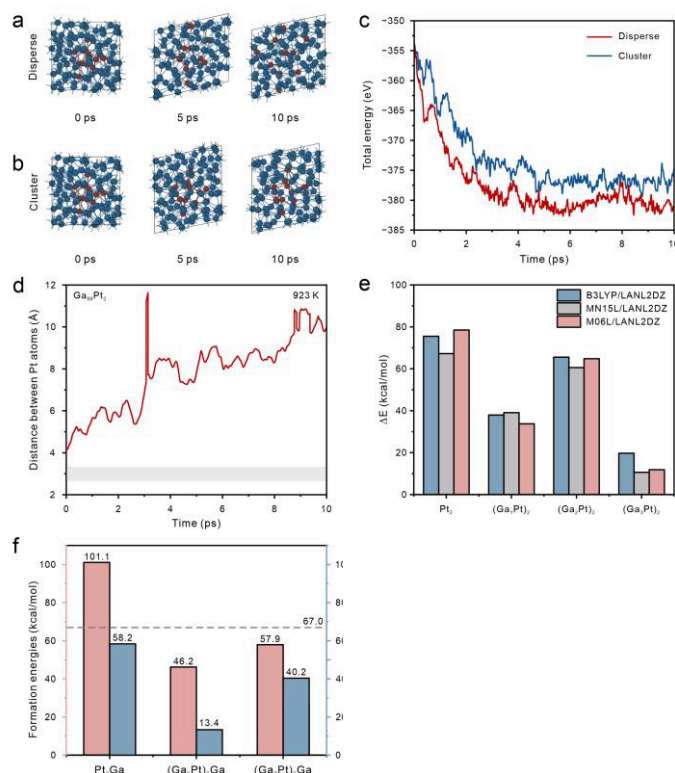
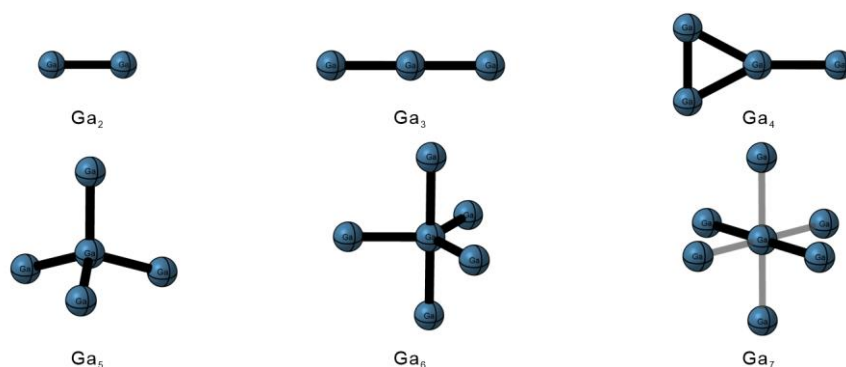


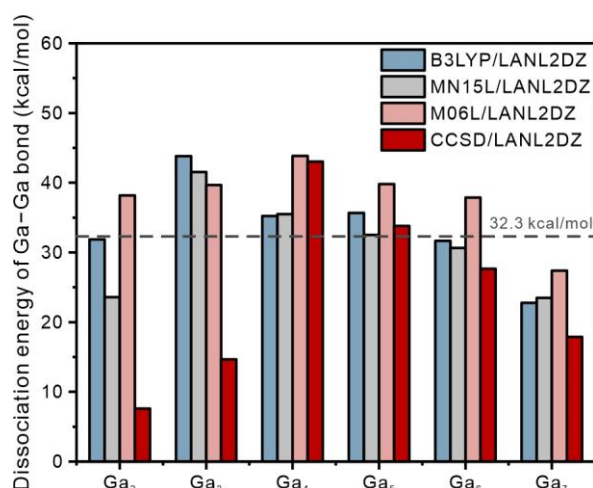
Fig. 2 | Theoretical calculations of heterogeneous active metal atoms monodispersed in liquid Ga. **a, b** The extracted 0 ps, 5 ps, and 10 ps snapshots from AIMD trajectories of both dispersed (**a**) and clustered (**b**) Pt_{10} in liquid Ga. **c** Total energy versus time curves for the $Ga_{90}Pt_{10}$ system containing clustered Pt_{10} and dispersed Pt_{10} . **d** Time evolution of Pt–Pt interatomic distance in the $Ga_{98}Pt_2$ system under NVT ensemble simulation. **e** The calculated bond energy difference ΔE_b using different computational methods (B3LYP, MN15L, and M06L) for the Pt_2 , $(Ga_1Pt)_2$, $(Ga_2Pt)_2$, and $(Ga_3Pt)_2$ molecular clusters. **f** The formation energies and energy barriers associated with Ga atom insertion into Pt–Pt bonds in Pt_2Ga , $(Ga_6Pt)_2Ga$, and $(Ga_7Pt)_2Ga$ molecular systems (the rough estimation suggests that the reaction can overcome an energy barrier of 67 kcal mol^{−1} at 923 K²⁷).

B3LYP/LANL2DZ, MN15L/ LANL2DZ⁵ and M06L/ LANL2DZ methods to calculate the bond energy difference ΔE_b ($\Delta E_b = E_{(Ga_nPt)_2(Pt-Pt)} + \bar{E}_{Ga_n(Ga-Ga)} - 2E_{Ga_{n+1}Pt(Ga-Pt)}$) only for the Pt₂, (Ga₁Pt)₂, (Ga₂Pt)₂, and (Ga₃Pt)₂ molecular clusters. Here, $\bar{E}_{Ga_n(Ga-Ga)}$ represents the average Ga–Ga bond energy in the Ga_n molecular clusters, $E_{(Ga_nPt)_2(Pt-Pt)}$ denotes the Pt–Pt bond energy in the (Ga_nPt)₂ molecular clusters, and $E_{Ga_{n+1}Pt(Ga-Pt)}$ refers to the Ga–Pt bond energy in the Ga_{n+1}Pt molecular clusters. All bond energy calculations were corrected for basis set superposition error (BSSE).

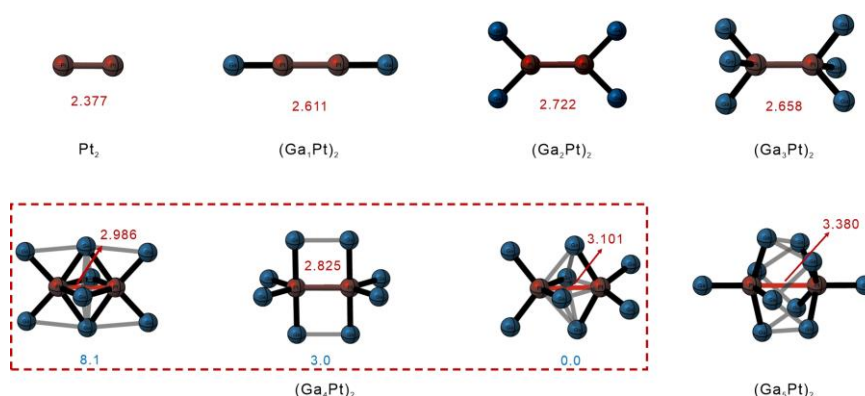
Page 13–18 in revised supplementary information:



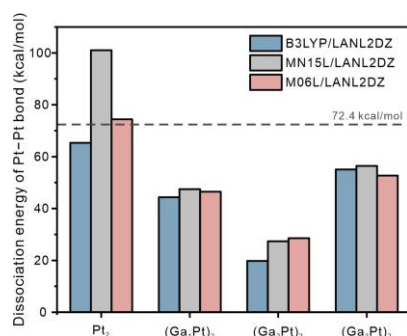
Supplementary Fig. 6 | The optimized structures structures of the Ga₂, Ga₃, Ga₄, Ga₅, Ga₆, and Ga₇Pt clusters.



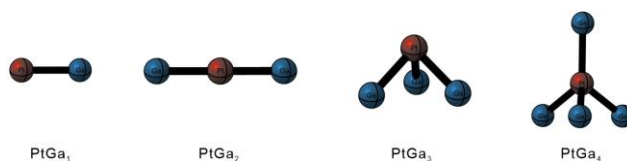
Supplementary Fig. 7 | The calculated dissociation energies of Ga–Ga bond in Ga₂, Ga₃, Ga₄, Ga₅, Ga₆, and Ga₇ clusters obtained using the B3LYP, MN15L, M06L, and CCSD methods. The Ga–Ga bond energies in the Ga₂ molecule calculated using the B3LYP, MN15L, M06L, and CCSD methods are 31.9, 23.6, 38.2, and 7.6 kcal mol^{−1}, respectively. Among these, the B3LYP result (31.9 kcal mol^{−1}) is closest to the experimental value⁹ of 32.3 kcal mol^{−1}, while the CCSD result deviates significantly from the experimental value, clearly indicating that the CCSD method is not suitable for simulating Ga–Pt systems. Furthermore, the average Ga–Ga bond energies in Ga₂, Ga₃, Ga₄, Ga₅, Ga₆, and Ga₇ clusters calculated using B3LYP, MN15L, and M06L methods are 33.5, 31.2, and 37.8 kcal mol^{−1}, respectively, demonstrating consistency among these three methods.



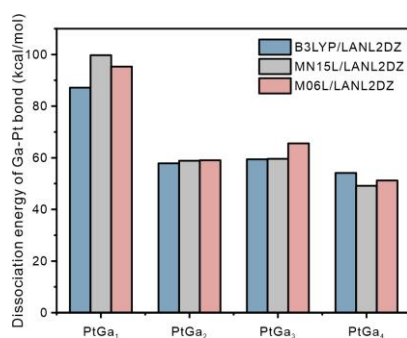
Supplementary Fig. 8 | The optimized structures of the molecular systems Pt₂, (Ga₁Pt)₂, (Ga₂Pt)₂, (Ga₃Pt)₂, (Ga₄Pt)₂, and (Ga₅Pt)₂. The red numbers denote the Pt–Pt bond lengths in angstroms (Å), and the blue numbers denote the relative energies (in kcal mol^{−1}) of the three isomers of (Ga₄Pt)₂. Based on the optimized geometries, the Pt–Pt bond lengths in the stable structures of (Ga₄Pt)₂ and (Ga₅Pt)₂ are 3.101 Å and 3.380 Å, respectively. These values significantly exceed 1.2 times the Pt–Pt bond length of 2.377 Å in Pt₂, indicating that the Pt–Pt bonds in the stable configurations of (Ga₄Pt)₂ and (Ga₅Pt)₂ can be considered effectively broken. Therefore, we calculated the bond energy difference ΔE_b only for the Pt₂, (Ga₁Pt)₂, (Ga₂Pt)₂, and (Ga₃Pt)₂ molecular clusters.



Supplementary Fig. 9 | The calculated dissociation energies of Pt–Pt bond in Pt₂, (Ga₁Pt)₂, (Ga₂Pt)₂, and (Ga₃Pt)₂ clusters obtained using the B3LYP, MN15L, M06L methods. the Pt–Pt bond energies in (Ga₁Pt)₂, (Ga₂Pt)₂, and (Ga₃Pt)₂ clusters calculated using B3LYP, MN15L, and M06L methods are similar, while the Pt–Pt bond energies in the Pt₂ molecule calculated with these methods are 65.3, 101.0, and 74.4 kcal mol^{−1}, respectively. Among these, the B3LYP and M06L results are close to the experimental value of 72.4 kcal mol^{−1}^[ref10].



Supplementary Fig. 10 | The optimized structures of the molecular systems PtGa₁, PtGa₂, PtGa₃, and PtGa₄.



Supplementary Fig. 11 | The calculated dissociation energies of Ga–Pt bond in PtGa₁, PtGa₂, PtGa₃, and PtGa₄ clusters obtained using the B3LYP, MN15L, M06L methods.

Comment 2: On the other hand, the B3LYP functional is known to be less accurate for transition metal dimers due to multi-configurational effects (see relevant references: Alvaro Posada-Borbón, Alvaro Posada-Amarillas, *Theoretical DFT study of homonuclear and binary transition-metal dimers, Chemical Physics Letters*, 2015; Haoyu S. Yu, Xiao He, Donald G. Truhlar, MN15-L: A New Local Exchange-Correlation Functional for Kohn–Sham Density Functional Theory with Broad Accuracy for Atoms, Molecules, and Solids, *J. Chem. Theory Comput.* 2016.) Given this, are the results in Fig. 2(e) sufficiently robust?

Author reply: Thank you for your kind reminding. The electronic structures of transition metal dimers (e.g., Cr₂, Mn₂, Fe₂, Co₂, Ni₂, Cu₂) exhibit considerable complexity due to the presence of multiple low-lying electron configurations. Consequently, the ground-state wave functions of these systems cannot be adequately represented by a single Slater determinant, rendering single-reference methods insufficient. These systems are characterized by pronounced strong electron correlation and substantive multiconfigurational character. For their accurate theoretical description, it is essential to employ multireference approaches, such as CASSCF/CASPT2 or MRCI. Though density functional theory (DFT) lacks a priori accuracy guarantees, its computational cost is significantly lower than that of multireference methods. Moreover, DFT methods have been widely applied to transition metal molecular systems for predicting molecular structures, vibrational frequencies, absorption spectra, transition-state structures, electron affinities, as well as thermochemical data (Cramer, C. J., Truhlar, D. G. Density functional theory for transition metals and transition metal chemistry. *Phys. Chem. Chem. Phys.* **2009**, *11*, 10757). Among the various DFT methods, the B3LYP method is a widely used approach (Lee, C., Yang, W., Parr, R. G. Development of the Colle-Salvetti correlation-energy formula into a functional of the electron density. *Phys. Rev. B* **1988**, *37*, 785). Moreover, in the reference you mentioned, “Posada-Borbón, A., Posada-Amarillas, A. Theoretical DFT study of homonuclear and binary transition-metal dimers. *Chem. Phys. Lett.* **2015**, *618*, 66” the authors point out that the B3LYP method performs well in describing experimental dissociation energy values for Pd–Au and Pt–Ni dimers, and the calculated equilibrium bond length r (Å) and harmonic frequency ω (cm⁻¹) for the Pt₂ dimer are 2.382 and 229.37, respectively, which are also in close agreement with the experimental values of 2.333 and 222.46. Additionally, since Ga is a main group metal element, the B3LYP computational method is entirely applicable. Therefore, the B3LYP method can, to some extent, be used for the Ga–Pt binary system we intend to study. Meanwhile, considering the paper you recommended, “Yu, H. S., He, X., Truhlar, D. G. MN15-L: A new local exchange-correlation functional for Kohn–Sham density functional theory with broad accuracy for atoms, molecules, and solids. *J. Chem. Theory Comput.* **2016**, *12*, 1280” we will employ hybrid functional (B3LYP), meta-GGA (MN15L and M06L), and coupled-cluster methods (CCSD(T)) to calculate the dissociation energies of different Ga–Pt systems.

First, as shown in Fig. R4, the Ga–Ga bond energies in the Ga₂ molecule calculated using the B3LYP, MN15L, M06L, and CCSD methods are 31.9, 23.6, 38.2, and 7.6 kcal mol⁻¹, respectively. Among these, the B3LYP result (31.9 kcal mol⁻¹) is closest to the experimental value of 32.3 kcal mol⁻¹ (Himmel, H. J., Gaertner, B. Characterization of Isolated Ga₂ Molecules by Resonance Raman Spectroscopy and Variations of Ga–Ga Bonding. *Chem. Eur. J.* **2004**, *10*, 5936), while the CCSD result deviates significantly from the experimental value, clearly indicating that the CCSD method is not suitable for simulating Ga–Pt systems. Furthermore, the average Ga–Ga bond energies in Ga₂,

Ga₃, Ga₄, Ga₅, Ga₆, and Ga₇ clusters calculated using B3LYP, MN15L, and M06L methods are 33.5, 31.2, and 37.8 kcal mol⁻¹, respectively, demonstrating consistency among these three methods. Additionally, as shown in Fig. R5, the Pt–Pt bond energies in (Ga₁Pt)₂, (Ga₂Pt)₂, and (Ga₃Pt)₂ clusters calculated using B3LYP, MN15L, and M06L methods are similar, while the Pt–Pt bond energies in the Pt₂ molecule calculated with these methods are 65.3, 101.0, and 74.4 kcal mol⁻¹, respectively. Among these, the B3LYP and M06L results are close to the experimental value of 72.4 kcal mol⁻¹ (Taylor, S., Lemire, G. W., Hamrick, Y. M., et al. Resonant two-photon ionization spectroscopy of jet-cooled Pt₂. *J. Chem. Phys.* **1988**, 89, 5517). Finally, as shown in Fig. R6, the Ga–Pt bond energies in PtGa₁, PtGa₂, PtGa₃, and PtGa₄ clusters calculated using B3LYP, MN15L, and M06L methods are also similar. **Overall, among the three methods (B3LYP, MN15L, and M06L), the B3LYP method stands out as particularly effective in accurately simulating the intermolecular interaction energies within Ga–Pt binary metal systems.** Therefore, the relevant results calculated using the B3LYP method are reliable.

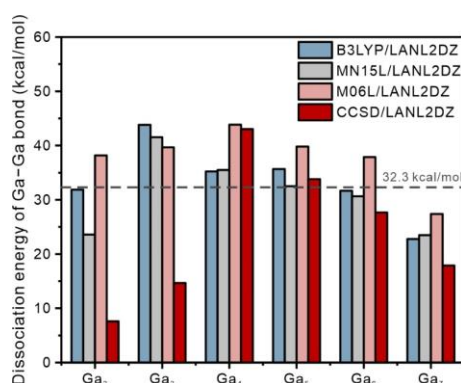


Fig. R4 | The calculated dissociation energies of the Ga–Ga bond in Ga₂, Ga₃, Ga₄, Ga₅, Ga₆, and Ga₇ clusters. The calculated dissociation energies were obtained using the B3LYP, MN15L, M06L, and CCSD methods. The dashed line indicates that the dissociation energy of the Ga–Ga bond in Ga₂ is 32.3 kcal mol⁻¹.

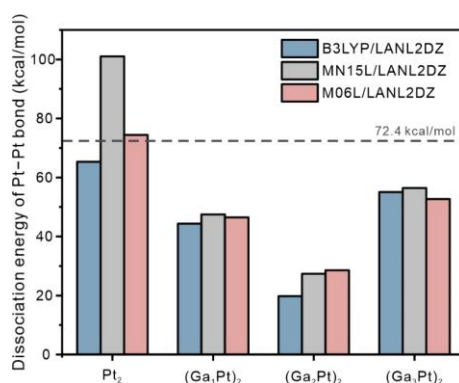


Fig. R5 | The calculated dissociation energies of the Pt–Pt bond in Pt₂, (Ga₁Pt)₂, (Ga₂Pt)₂, and (Ga₃Pt)₂ clusters. The calculated dissociation energies were obtained using the B3LYP, MN15L, and M06L methods. The dashed line indicates that the dissociation energy of the Ga–Ga bond in Ga₂ is 72.4 kcal mol⁻¹.

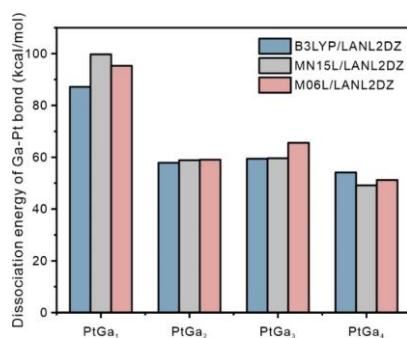


Fig. R6 | The calculated dissociation energies of the Ga–Pt bond in PtGa₁, PtGa₂, PtGa₃, and PtGa₄ clusters. The calculated dissociation energies were obtained using the B3LYP, MN15L, and M06L methods.

Changes: We employed three computational methods (B3LYP, MN15L, and M06L) to demonstrate the reliability of selecting the B3LYP method for calculations. The changes are marked in red in the revised manuscript (Supplementary Fig. 6–11).

Comment 3: *Additionally, what is the turnover frequency (TOF) normalized by the mass of Pt? This information is crucial for evaluating the atomic efficiency and cost-effectiveness of the catalyst, given the high price of platinum?*

Author reply: Thank you very much for your valuable suggestions. We agree that the mass-based TOF is an indispensable parameter for assessing the economic viability of Pt-based catalysts. According to your suggestion, we supplemented the corresponding experiments and calculated the TOF value normalized by the mass of Pt. Furthermore, we analyze the cost of Pt SACs@Ga/ZSM-5, Pt particles/ZSM-5 catalyst, and pure Pt to evaluate the commercial potential of our catalyst.

(1) TOF Calculation. The turnover frequency (TOF) for ethylene production was calculated using the following equation:

$$TOF_{C_2H_4} = \frac{F_{C_2H_6} \times Conv_{C_2H_6} \times Sel_{C_2H_4}}{m_{cat} \times \omega_{Pt} \times D_{Pt}}$$

where $F_{C_2H_6}$, $Conv_{C_2H_6}$ and $Sel_{C_2H_4}$ represent the flow rate and conversion of ethane, and the selectivity of ethylene, respectively. And m_{cat} represents the total mass of the catalyst. ω_{Pt} represents the total mass fraction of active metal Pt in the catalyst, which is determined by ICP-OES. D_{Pt} represent the dispersion of the active Pt on the surface of the catalyst. Based on the measurement of CO pulse adsorption, D_{Pt} is approximately 13.96%. Based on these data, the calculated TOF for ethylene over our catalyst is 3.69 h⁻¹.

(2) Cost analyses. Based on the calculation methods of the reference, a comprehensive cost

analysis was conducted to evaluate the commercial potential of our catalyst (*Sci. Adv.* **2025**, *11*, eadt9037). The total cost to produce 10 kg of three different catalysts was determined using current global metal prices and supplier quotations, with the calculation methodology detailed below. As summarized in Tab. R1 and illustrated in Fig. R7, our catalyst achieves a cost reduction relative to the pure Pt catalyst. Notably, it also undercuts the cost of a conventional Pt particles/ZSM-5 catalyst by an additional 9.2%, while delivering enhanced selectivity and conversion.

Tab. R1 | The total cost to produce 10 kg of three different catalysts. The prices of Pt and Ga are from the website: <https://www.dailymetalprice.com/metalpricecharts.php>, and the price of ZSM-5 is from Aladdin’s selling price (https://www.aladdin-e.com/zh_cn/m196697.html#citations-of-this-product).

Type of catalyst	Mass of catalyst	Materials cost (US\$)			Total cost (US\$)
		Pt	ZSM-5	Ga	
Pure Pt		607,607.1	/	/	607,607.1
Pt particles/ZSM-5	10 kg	1,822.8	3, 379.1	/	5,201.9
Pt SACs@Ga/ZSM-5		1,822.8	1,830.2	1,067.3	4,720.3

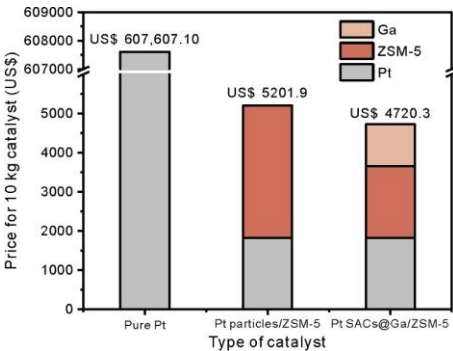


Fig. R7 | The total cost to produce 10 kg of three different catalysts. The prices of Pt and Ga are from the website: <https://www.dailymetalprice.com/metalpricecharts.php>, and the price of ZSM-5 is from Aladdin's selling price (https://www.aladdin-e.com/zh_cn/m196697.html#citations-of-this-product)

Cost analysis details:

1) Pure Pt

Pt price = US\$ 60,760.71/kg (as of Dec 09, 2025)

10 kg of catalyst = 10 kg × US\$ 60,760.71/kg = US\$ 607,607.1

Total material cost = US\$ 607,607.1

Note: The average price of Pt is approximately US\$ 1701.30 oz = US\$ 60,760.71 kg (as of Dec 09, 2025)

<https://www.dailymetalprice.com/metalpricecharts.php>

2) Pt particles/ZSM-5

The amount of Pt used in Pt particles/ZSM-5 and Pt SACs@Ga/ZSM-5 catalyst is the same.

Cost Calculation for Pt particles/ZSM-5 Catalyst:

Cost of Pt price = $0.3\% \times 10 \text{ kg} \times \text{US\$ } 60,760.71/\text{kg} = \text{US\$ } 1,822.8$ (as of Dec 09, 2025)

Cost of ZSM-5 molecular sieve = $99.7\% \times 10 \text{ kg} \times \text{RMB¥ } 2398/\text{kg} \div 7.0753 = \text{US\$ } 3,379.1$ (see note below)

Total material cost = $\text{US\$ } 1,822.8 \text{ (Pt)} + \text{US\$ } 3,379.1 \text{ (ZSM-5)} = \text{US\$ } 5,201.9$

Note: Prices are approximate and may vary based on supplier, purity, and purchase quantity. Bulk quantity purchase will reduce the cost.

i) ZSM-5 molecular sieve is a specialized chemical, and Aladdin offers ZSM-5 molecular sieve at RMB¥ 1199 for 500 grams, equating to RMB¥ 2398/kg (as of Dec 09, 2025)

https://www.aladdin-e.com/zh_cn/m196697.html#citations-of-this-product

ii) According to the exchange rate announced by the Bank of China, the exchange rate between the US dollar and the Chinese yuan is: $\text{RMB¥ } 7.0753 \approx \text{US\$ } 1$ (as of Dec 10, 2025)

<https://www.chinamoney.com.cn/chinese/sddshl/>

3) Pt SACs@Ga/ZSM-5

This catalyst contains 0.3 wt% Pt (The atomic ratio of the liquid catalyst is Ga : Pt = 99.8 : 0.2)

Cost Calculation for Pt SACs@Ga/ZSM-5 Catalyst:

Cost of Pt price = $0.3\% \times 10 \text{ kg} \times \text{US\$ } 60,760.71/\text{kg} = \text{US\$ } 1,822.8$

Cost of Ga price = $45.7\% \times 10 \text{ kg} \times \text{US\$ } 233.54/\text{kg} = \text{US\$ } 1,067.3$

Cost of ZSM-5 molecular sieve = $54.0\% \times 10 \text{ kg} \times \text{RMB¥ } 2398/\text{kg} \div 7.0753 = \text{US\$ } 1,830.2$ (see note below)

Total material cost = $\text{US\$ } 1,822.8 \text{ (Pt)} + \text{US\$ } 1,067.3 \text{ (Ga)} + \text{US\$ } 1,830.2 \text{ (ZSM-5)} = \text{US\$ } 4,720.3$

Note: The average price of Ga is approximately US \$233.54/kg (as of Dec 09, 2025)

<https://www.dailymetalprice.com/metalpricecharts.php>